



Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

A detailed combined experimental and theoretical study on dimethyl ether/propane blended oxidation

Enoch E. Dames^{a,*}, Andrew S. Rosen^a, Bryan W. Weber^b, Connie W. Gao^a, Chih-Jen Sung^b, William H. Green^a

^a Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA, USA

^b Department of Mechanical Engineering, University of Connecticut, Storrs, CT, USA

ARTICLE INFO

Article history:

Received 4 December 2015

Revised 18 February 2016

Accepted 18 February 2016

Available online xxx

Keywords:

Dimethyl ether

Propane

Kinetic model

Rapid compression machine

Low temperature combustion

ABSTRACT

In this paper, a binary fuel model for dimethyl ether (DME) and propane is developed, with a focus on engine-relevant conditions (10–50 atm and 550–2000 K). New rapid compression machine (RCM) data are obtained for the purpose of further validating the binary fuel model, identifying reactions important to low-temperature propane and DME oxidation, and understanding the ignition-promoting effect of DME on propane. It is found that the simulated RCM data for DME/propane mixtures is very sensitive to the rates of $C_3H_8 + OH$, which acts as a radical sink relative to DME oxidation, especially at high relative DME concentrations. New rate evaluations are conducted for the reactions of $C_3H_8 + OH = products$ as well as the self-reaction of methoxymethyl peroxy (in competition with $RO_2 = QOOH$ isomerization) of $2CH_3OCH_2O_2 = products$. Accurate phenomenological rate constants, $k(T, P)$, are computed by RRKM/ME methods (with energies obtained at the CCSD(T)-F12a/cc-pVTZ-F12 level of theory) for several radical intermediates relevant to DME. The model developed in this paper (120 species and 711 reactions) performs well against the experimental targets tested here and is suitable for use over a wide range of conditions. In addition, the reaction mechanism generator software RMG is used to explore cross-reactions between propane and DME radical intermediates. These cross-reactions did not have a significant effect on simulations of the conditions modeled in this paper, suggesting that kinetic models for high- and low-reactivity binary fuel mixtures may be assembled from addition of their corresponding submodels and a small molecule foundation model.

© 2016 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Dimethyl ether (DME, CH_3OCH_3) is widely known to be a good transportation fuel, in part because it can act as a drop-in replacement for diesel while reducing soot, carbon monoxide (CO), and unburned hydrocarbon emissions [1]. As a fuel additive to conventional fossil fuel in internal combustion engines, DME has the potential to greatly reduce particulate (soot) formation [2]. Fuel additives have been studied for their potential to control ignition timing of homogeneous-charge compression-ignition (HCCI) engines, stratified-charge compression-ignition (SCCI) engines, and reactivity-controlled compression-ignition (RCCI) engines. In combination with intelligent fuel mixture and composition design, these advanced engines offer the prospect of cleaner burning and higher thermal efficiencies [3–6]. Many of these future engines may rely on dual fuel systems or those with adjustable

concentrations of fuel additives that advance combustion [7]. However, these same engines will also rely more heavily on fundamental knowledge of the underlying combustion chemistry that drives the low-temperature autoignition of fuel mixtures with both high- and low-reactivity components.

The promoting effect of DME on fuels such as methane (CH_4) under conditions of both low- and high-temperature combustion has been studied in some detail [8,9]. Under high-temperature regimes, the rapid build-up of radicals formed from DME promotes methane ignition, as noted by Chen et al. [8]. Although there is a nonlinear ignition enhancement, the laminar flame speeds of DME/methane mixtures are linearly proportional to the DME fraction [8]. The ignition promoting effect of DME on high-octane fuels is of particular importance for controlling ignition timing in internal combustion engines. Fundamental kinetic studies of fuel blends and quantification of ignition enhancement may therefore be useful in modeling, optimizing, and ultimately developing more advanced and cleaner combustion engines.

Dual fuel combustion embodies a conceptually novel approach to internal combustion engine technology, relying on mixtures of

* Corresponding author.

E-mail address: edames@mit.edu, enoch.dames@gmail.com (E.E. Dames).

<http://dx.doi.org/10.1016/j.combustflame.2016.02.021>

0010-2180/© 2016 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Table 1

Recent experimental studies of DME combustion with their corresponding experimental conditions, including temperature, pressure, equivalence ratio (ϕ), diluent, and residence time (t_{res}). Adapted from Rodriguez et al. [19].

Type of experiment	Experimental conditions	Reference
Jet-stirred reactor	$T = 800\text{--}1300\text{ K}$; $P = 101\text{--}1013\text{ kPa}$; $\phi = 0.2\text{--}2$ in N_2 ; $t_{res} = 0.1\text{--}1\text{ s}$ $T = 550\text{--}1100\text{ K}$; $P = 1013\text{ kPa}$; $\phi = 0.2\text{--}1.0$ in N_2 ; $t_{res} = 1\text{ s}$ $T = 540\text{--}850\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.5\text{--}2$ in N_2 ; $t_{res} = 1.5\text{ s}$ $T = 550\text{--}1100\text{ K}$; $P = 107\text{ kPa}$; $\phi = 0.25\text{--}2$ in He , $t_{res} = 2\text{ s}$	Dagaut et al. 1996 [32]* Dagaut et al. 1998 [21]* Le Tan et al. 2015 [33] Rodriguez et al. 2015 [19]*†
Plug flow reactor	$T = 600\text{--}1500\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.01\text{--}\infty$ in air $T = 1080\text{--}1086\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.5$ in N_2 ; $t_{res} = 0.1\text{ s}$ $T = 550\text{--}855\text{ K}$; $P = 1216\text{--}1825\text{ kPa}$; $\phi = 0.7\text{--}4.2$ in N_2 ; $t_{res} = 1.8\text{ s}$ $T = 513\text{--}973\text{ K}$; $P = 101\text{ kPa}$; 340 ppm in 10% O_2 ; $t_{res} = 2\text{--}4\text{ s}$ $T = 490\text{--}750\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.6$ in He ; $t_{res} = 107\text{ s}$ $T = 500\text{--}1200\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.8\text{--}1.2$ in Ar $T = 739\text{--}902\text{ K}$; $P = 200\text{--}400\text{ kPa}$; $\phi = 0.225\text{--}0.675$ in N_2 $T = 555\text{--}585\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.835$ in air $T = 500\text{--}1150\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.2\text{--}1.6$ in He/Ar ; $t_{res} = 0.19\text{--}2\text{ s}$ $T = 500\text{--}1150\text{ K}$; $P = 101\text{ kPa}$; $\phi = 1$ in Ar ; $t_{res} = 1.5\text{ s}$	Alzueta et al. 1999 [34] Fischer et al. 2000 [17] Curran et al. 2000 [18] Liu et al. 2001 [35] Guo et al. 2013 [36] Herrmann et al. 2013, 2014 [37][38] Schönborn et al. 2014 [39] Wada et al. 2013 [40] Kurimoto et al. 2015 [41] Wang et al. 2015 [42]
Shock tube	$T = 650\text{--}1300\text{ K}$; $P = 1317\text{--}4053\text{ kPa}$; $\phi = 1$ in air $T = 1200\text{--}1600\text{ K}$; $P = 350\text{ kPa}$; $\phi = 0.5\text{--}2$ in Ar $T = 1175\text{--}1900\text{ K}$; $P = 161\text{--}666\text{ kPa}$; $\phi = 0.5\text{--}3$ in Ar $T = 1134\text{--}2015\text{ K}$; $P = 101\text{--}1010\text{ kPa}$; $\phi = 1$ in Ar $T = 1100\text{--}1500\text{ K}$; $P = 2000\text{ kPa}$; $\phi = 0.5\text{--}2$ in Ar $T = 1200\text{--}1600\text{ K}$; $P = 120\text{--}530\text{ kPa}$; $\phi = 1$ in Ar $T = 697\text{--}1239\text{ K}$; $P = 2200\text{--}2300\text{ kPa}$; $\phi = 0.5\text{--}1.5$ in air/N_2 $T = 1300\text{--}1600\text{ K}$; $P = 152\text{ kPa}$; 0.5–2% DME in Ar $T = 900\text{--}1700\text{ K}$; $P = 122\text{--}1013\text{ kPa}$; $\phi = 0.5\text{--}2$ in H_2/Ar $T = 400\text{--}1160\text{ K}$; $P = 1111\text{--}3030\text{ kPa}$; $\phi = 0.3\text{--}2$ in air $T = 1000\text{--}1600\text{ K}$; $P = 122\text{--}2030\text{ kPa}$; $\phi = 0.5\text{--}2$ in Ar	Pfahl et al. 1996 [22]* Dagaut et al. 1998 [21]* Cook et al. 2009 [43]* Tang et al. 2012 [44]* Hu et al. 2013 [29]† Hu et al. 2013 [29] Li et al. 2013 [23]† Pyun et al. 2013 [45]* Pan et al. 2014 [46]* Burke et al. 2015 [9]* Pan et al. 2015 [47]
RCM	$T = 615\text{--}735\text{ K}$; $P = 1010\text{--}2010\text{ kPa}$; $\phi = 0.43\text{--}1.5$ in N_2 $T = 630\text{--}1250\text{ K}$; $P = 1111\text{--}3030\text{ kPa}$; $\phi = 0.3\text{--}2$ in air	Mittal et al. 2008 [15]*† Burke et al. 2015 [9]*
Burner	premixed flame; $P = 4\text{ kPa}$; $\phi = 0.98\text{--}1.2$ in Ar premixed flame; $P = 101\text{ kPa}$; $\phi = 0.67\text{--}1.49$ in air premixed flame; $P = 2.67\text{--}4\text{ kPa}$; $\phi = 1.2\text{--}1.68$ in Ar premixed flame; $P = 4\text{ kPa}$; $\phi = 0.5\text{--}2$ in Ar premixed flame; $P = 4\text{ kPa}$; $\phi = 1$ in Ar premixed flame; $P = 5\text{ kPa}$; $\phi = 1.63$ in Ar/CO_2 Single jet-wall stagnation flame; $T = 298\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.7\text{--}1.7$ in air Counterflow flame; $T = 298\text{ K}$; $P = 101\text{ kPa}$; $\phi = 0.7\text{--}1.7$ in air	McIlroy et al. 2000 [48] Kaiser et al. 2000 [49] Cool et al. 2007 [50] Wang et al. 2008 [51] Xu et al. 2011 [52] Liu et al. 2013 [12]* Zhao et al. 2008 [20] Wang et al. 2009 [53]
Spherical bomb	$T = 295\text{ K}$; $P = 100\text{ kPa}$; $\phi = 0.7\text{--}1.7$ in air $T = 298\text{ K}$; $P = 200\text{--}1000\text{ kPa}$; $\phi = 0.6\text{--}1.7$ in air $T = 298\text{ K}$; $P = 80\text{--}150\text{ kPa}$; $\phi = 0.7\text{--}1.8$ in air $T = 293\text{ K}$; $P = 97\text{ kPa}$; $\phi = 0.7\text{--}1.8$ in air $T = 298\text{ K}$; $P = 100\text{--}1000\text{ kPa}$; $\phi = 0.7\text{--}1.6$ in air	Daly et al. 2001 [54] Qin and Ju 2005 [55] Huang et al. 2007 [56] Chen et al. 2009 [57] De Vries et al. 2011 [58]

* See the Supplementary material for comparisons with the model of the present work.

† See main text for comparisons with the model of the present work.

high-octane fuels and high-cetane fuels as well as their synergistic combustion characteristics in order to achieve high thermal efficiencies. These engines can be made from retrofitted diesel engines with modified intake manifolds to allow for fuel vapor mixing into the engine piston. For example, recent work [4] has identified a mixture of 20% DME with 30% propane (by mass) in concert with diesel fuel combustion, resulting in over 50% brake thermal efficiency (BTE) in a modified rail diesel engine, compared to a baseline BTE of 37%. The authors found that propane substitution delayed DME's early autoignition and shifted the combustion process closer to top dead center [4]. In addition, DME has been investigated as an additive for ignition control in HCCI engines [1,7]. The combustion kinetics of fuel blends has also received wide attention in efforts to understand the oxidation and pyrolysis of primary reference fuels [10]. As a result, studies of DME combustion abound in the literature. Recent work (see Table 1) includes studies of premixed and non-premixed flames, homogeneous ignition experiments, and studies in homogeneous flow reactors.

The uncertainties in low-temperature (400–900 K) DME oxidation kinetics primarily involve the first and second O_2 additions (typically denoted $R + O_2 = RO_2$ and $QOOH + O_2 = O_2QOOH$, respectively) to radical derivatives of DME as well as the fate of

ketohydroperoxide, HO_2CH_2OCHO (hydroperoxymethylformate), all of which can serve to generate OH radicals, thereby promoting heat release and ignition. Tomlin et al. [11] have clearly illustrated the need for further work in this area by pointing out that several literature models such as [12–14] fail to accurately reproduce earlier rapid compression machine (RCM) ignition delay time measurements of DME [15]. In particular, Tomlin et al. [11] highlighted the need for accurate pressure-dependent descriptions for both first and second O_2 additions to DME radical derivatives and subsequent OH forming pathways. Thus, a number of more recent studies have been published that include detailed rate theory calculations for these reactions. Notably, Eskola et al. [16] performed a combined theoretical and experimental study for both first and second O_2 additions. The subsequent corresponding $CH_3OCH_2O_2$ (RO_2) = CH_2OCH_2OOH ($QOOH$) isomerization step is determined to be faster than those used in previous models [17,18], suggesting that the O_2 addition to $QOOH$ may play a more important role than previously thought [16]. The rate theory calculations performed by Eskola et al. [16] resulted in a total of nine phenomenological rates, including four well-skipping reactions. The rate of the $RO_2 = QOOH$ isomerization is near its high-pressure limit at 1 bar, while the rate of the important $QOOH$ decomposition to $2CH_2O + OH$ is at its high-pressure limit at 10 bar. Rodriguez et al. [19] also

recently calculated high-pressure limit rates on a G4 potential energy surface for the reactions $\text{CH}_3\text{OCH}_2\text{O}_2 = \text{CH}_2\text{OCH}_2\text{OOH}$ and $\text{CH}_3\text{OCH}_2\text{O}_2 = \text{CH}_3\text{OCHO} + \text{OH}$. The same authors conducted atmospheric-pressure jet-stirred reactor (JSR) experiments for varying equivalence ratios, measuring a number of intermediate and product species profiles using gas chromatography and cavity ring-down spectroscopy. Rodriguez et al. [19] noted that the use of both the DME models of Zhao et al. [20] and Burke et al. [9] overestimate the DME reactivity in the low-temperature region. Thus, Rodriguez et al. [19] assembled a new model which satisfactorily reproduces their measured species profiles in addition to a number of higher-pressure experiments, including other JSR data at 10 atm [21], plug flow reactor (PFR) data at 12.5 atm [18], and shock tube data ranging from 12 to 40 bar [9,22,23]. Finally, Burke et al. [9] computed pressure-dependent rates for the unimolecular decomposition of CH_3OCH_2 , by performing Quantum-Rice-Ramsperger-Kassel (QRRK) calculations with a modified strong-collisional model for energy transfer, and energies obtained at the CBS-QB3 level of theory.

Propane (C_3H_8) has also been studied extensively, and its combustion has been reviewed in several studies, including those by Simmie [24] and Westbrook and Dryer [25]. Thus, only select previous works are noted briefly, which are relevant to this paper. Of particular interest to this paper is the study by Gallagher et al. [26], who measured pure propane overall ignition delay times in a rapid compression machine for a variable equivalence ratio. In addition, the low-temperature chain branching pathways (involving the first and second O_2 additions) have been studied using high-level electronic structure and rate theory methods by Goldsmith et al. [27]. Although not tested against global experimental measurements (e.g., ignition delay times) in a detailed kinetic model, individually computed rates were compared with those available in the literature. More recently, Merchant et al. [28] conducted a study on the first-stage ignition of propane, using the elementary rates computed by Goldsmith et al. [27]. With a focus on deriving simple analytical solutions that can reliably predict first-stage ignition delay times, Merchant et al. [28] discussed in depth the roles of both OH and HO_2 radicals on the low-temperature autoignition behavior of propane.

Fundamental combustion studies on DME/hydrocarbon mixtures have also been recently performed, albeit to a lesser extent than for each individual component. Hu et al. [29] have studied DME/propane autoignition at engine-relevant pressures (20 bar) by measuring ignition delay times behind reflected shock waves at high temperatures (1100–1500 K). Burke et al. [9] performed a combined experimental and modeling study of DME/ CH_4 mixtures between 600–1600 K and 7–41 atm for varying fuel equivalence ratios. The authors performed rate theory calculations to elucidate the pressure-dependent nature of the unimolecular decomposition of the CH_3OCH_2 radical, finding that this chemistry, in addition to pressure-dependent low-temperature DME chemistry adopted from Yamada et al. [30], resulted in improvements in model agreement with various experimental data compared to modeling with only pressure-independent expressions. Burke et al. [9] also pointed out the strong promoting effect that small amounts of DME can have on overall ignition delays of 20:80 blends (by mole) of DME: CH_4 . This promoting effect is attributed to the overall weaker C–H bonds of DME compared to that of methane.

Although reactions between dual fuel derivatives/intermediates may exist under engine conditions, no current model incorporating both DME and propane reaction kinetics includes these fuel intermediate cross-reactions. Most models of fuel blends do not contain any direct cross-reactions between the first intermediate radicals produced during oxidation/pyrolysis, although some exceptions do exist. In a model of gasoline surrogate components,

Mehl et al. [10] added cross-reactions of alkyl and alkyl-peroxy radicals to account for interactions of various fuel components in the mixture, although their importance in determining various combustion characteristics was not discussed.

Thus, the first objective of this paper is to investigate the capability of the Reaction Mechanism Generator (RMG) [31] to identify any reactions important to blended fuel combustion chemistry that do not exist in a pre-constructed base model from a simple concatenation of existing DME and propane mechanisms. Use of RMG in this manner can be especially valuable in the context of dual- or multi-component fuel mixtures given that many potentially important cross-reactions between fuel components may be readily missed or overlooked during manual (i.e., human) mechanism construction. The second objective of this paper is to collect new experimental data in a rapid compression machine for the autoignition of binary DME/propane mixtures under engine-relevant conditions (a pressure of 30 bar and temperatures of 600–900 K). The third objective of this study is to conduct rate calculations and critical evaluations for several reactions important to the autoignition of DME and propane. These improved rate calculations are used to assemble a comprehensive model for the combustion of binary DME/propane mixtures under engine-relevant conditions (initial pressures of 1–60 bar and initial temperatures of 300–1500 K). This model is validated by comparison to a wide range of experimental targets, including literature data and the new data for DME/propane binary mixtures collected in this study. Finally, we investigate the non-linear promoting effect of DME on propane autoignition.

2. Experimental methods

The present experiments are conducted in a heated RCM at the University of Connecticut. The specifications of this RCM have been detailed in [59]. Briefly, the RCM is a single-piston, pneumatically-driven, hydraulically-stopped arrangement with compression times near 30 ms. The end of compression (EOC) temperature and pressure conditions, T_c and P_c respectively, are independently changed by varying the compression ratio, initial pressure, and initial temperature of the experiments. The piston in the reaction chamber is machined with specially designed crevices to suppress the roll-up vortex effect and promote homogenous conditions during and after compression [60].

The primary diagnostic on the RCM is the in-cylinder pressure, measured by a Kistler 6125C transducer coupled to a Kistler 5010B charge amplifier. The voltage from the charge amplifier is recorded by a National Instruments 9215 analog input device connected to a cDAQ 9178 chassis. The voltage is sampled at a rate of 100 kHz by a LabView VI and processed by a Python package called UConnRCMPy developed as part of this paper; version 1.0.7 of UConnRCMPy was used in this study. The code for the data processing package is available on GitHub at <https://github.com/bryanwweber/UConnRCMPy>.

The compression stroke of the RCM brings the homogenous fuel/oxidizer mixture to the EOC conditions, and for suitable values of T_c and P_c , the mixture will ignite after some delay. For some combinations of mixture/pressure/temperature conditions, the ignition undergoes two stages. In these cases, the first-stage ignition delay (τ_1) is defined as the time from the EOC ($t = 0$) until the first peak in the time derivative of the pressure trace, while the overall ignition delay (τ) is defined as the time from the EOC until the global maximum of the time derivative of the pressure trace, as shown in Fig. 1. Cases undergoing single-stage ignition have their ignition delay defined the same as the overall ignition delay.

Each experimental condition is repeated at least five times to ensure repeatability of the data. As there is some random

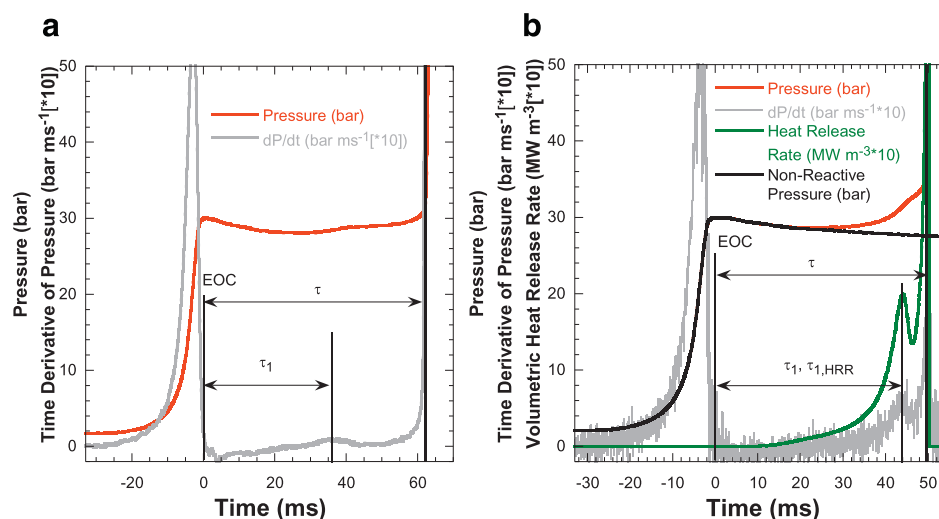


Fig. 1. Definitions of ignition delays, τ_1 and τ , in the RCM experiments and simulations. The conditions shown are for a $\phi = 2.0$ mixture of propane in O₂/N₂ air. (a) shows an experimental pressure trace with conditions: $P_0 = 1.71$ bar, $T_0 = 373$ K, $P_C = 30.0$ bar, $T_C = 759$ K. (b) shows a simulated pressure trace with conditions: $P_0 = 2.06$ bar, $T_0 = 373$ K, $P_C = 30.0$ bar, $T_C = 726$ K.

scatter present in the data, the standard deviation (σ) of the ignition delays from the runs at a particular condition is computed. In all cases, σ is less than 10% of the mean value of the ignition delay. Vertical error bars shown on plots of ignition delay measured in this study represent twice the standard deviation of the experiments at a given condition; if no error bars are shown for data reported in this study, the error bars are approximately the same vertical dimension as the data point.

In addition to the reactive experiments, non-reactive experiments are carried out to determine the influence of the machine-specific operating parameters on the experimental conditions. In these non-reactive experiments, O₂ in the oxidizer is replaced with N₂ to maintain a similar specific heat ratio but suppress oxidation reactions that lead to thermal runaway. If the pressure at the EOC of the non-reactive experiments matches that at the EOC of the reactive experiments, it is assumed that no substantial heat release has occurred during the compression stroke, and the temperature at the EOC (T_C) can be estimated by applying the adiabatic core hypothesis [61] and the isentropic relations between pressure and temperature during the compression stroke:

$$\ln\left(\frac{P_C}{P_0}\right) = \int_{T_0}^{T_C} \frac{\gamma}{\gamma - 1} \frac{dT}{T}$$

where P_0 is the initial pressure, T_0 is the initial temperature, γ is the temperature-dependent specific heat ratio, and the other variables are as defined previously. Due to the varying specific heat ratio, this formula is not integrated directly. Instead, the procedure described in Section 3.2.2 is used to model the compression stroke, and the simulated temperature at the EOC is taken as T_C . The uncertainty inherent in this calculation of T_C was estimated by Weber et al. [62] using a Monte Carlo method. Using the same method for the conditions of this experiment results in a typical uncertainty in the EOC temperature of $\pm 1\%$, similar to the values obtained in the work of Weber et al. [62]. Horizontal error bars are shown on the experimental data obtained in this study to represent this $\pm 1\%$ uncertainty.

The experiments in this paper are carried out for the stoichiometric equivalence ratio for a range of relative fractions of DME/C₃H₈ in the fuel blends, from pure DME to pure C₃H₈. The oxidizer for all the stoichiometric experiments is a mixture of oxygen and nitrogen in the molar ratio of 1:3.76 (i.e., air). Furthermore, two additional conditions at $\phi = 0.5$ and 2.0 are studied for

Table 2

Summary of RCM experimental conditions.

Molar proportions (purity of reactant)				Equivalence ratio (ϕ)
DME (99.8%)	C ₃ H ₈ (99.72%)	O ₂ (99.994%)	N ₂ (99.999%)	
1	0	3	11.28	1.0
3	1	14	52.64	1.0
1	1	8	30.08	1.0
1	3	18	67.68	1.0
1	9	48	180.48	1.0
0	1	5	18.80	1.0
0	1	10	13.80	0.5
0	1	2.5	21.30	2.0

pure C₃H₈, by holding the initial fuel mole fraction constant while adjusting the initial O₂ and N₂ mole fractions. All the experiments are conducted for an EOC pressure of $P_C = 30$ bar, and the EOC temperatures (T_C) range from 603 to 902 K. The mixture conditions are summarized in Table 2, and the complete list of experimental data points collected in this study is presented in the Supplementary material.

Mixtures are prepared in one of two mixing tanks, approximately 15 L or 17 L in volume. The larger tank is used for the reactive mixtures, while the smaller is used for the non-reactive mixtures as described above. Mixtures are made by first vacuuming the mixing tanks to an ultimate pressure less than 5 torr, and then filling each gas sequentially into the mixing tank to the desired partial pressure. The filling is done at room temperature. Typical total pressures in each tank after filling range from 1700 to 2500 torr. The pressure in the mixing tanks during filling is monitored by an Omega Engineering MMA type static pressure transducer. This same transducer is used to measure the initial pressure of the reactants prior to a given experiment. In addition, the temperatures of the mixing tanks and reaction chamber are monitored by several K-type thermocouples; these thermocouples are also used as the control feedback for the heaters. After filling is complete, the heaters are switched on, and the mixture is allowed 2 h to come to steady state. The mixing tanks are also equipped with magnetic stir bars to enhance mixing and ensure homogeneity of the reactants.

3. Computational methods

3.1. Rate theory calculations

Electronic energies of all species studied in this paper were determined at the CCSD(T)-F12a/cc-pVTZ-F12 [63–66] level of theory using Molpro [67]. Molecular geometries and force constants for species and saddle points were determined at the M08SO/MG3S [68, 69] level of theory, utilizing QChem 4.1 [70]. A computational grid with 75 radial points and 434 angular points per radial point was used in the optimization and frequency calculations for all species. Frequencies were scaled by the recommended value of 0.983 [71]. Loose internal degrees of freedom for relevant adducts and transition states (i.e., hindered rotors) were treated separately by performing relaxed potential energy scans about the bond defining the internal rotor; these calculations were performed in Gaussian 03 [72] at the BMK/6-311+G(d,p) level of theory [73]. Reduced internal moments of inertia for all internal rotors were estimated at the $\Gamma^{(2,3)}$ level as defined by East and Radom [74]. Cantherm [75], a utility of RMG, was used for all TST and RRKM/Master Equation calculations, which were performed in the region of 0.001–100 atm in a nitrogen bath gas and varying temperatures based on the system (further discussed below). The master equation for energy transfer was solved using the chemically significant eigenvalue method, which is documented elsewhere [31,76]. Eckart tunneling corrections [77] were applied to all relevant reactions. All information pertaining to the calculations in this paper can be found in the Supplementary material, including rate coefficients suitable for combustion modeling, in both Chebyshev and PLOG [78] formats.

The exponential down model for collisional energy transfer was adopted in this paper. A temperature-dependent formulation was used for the average downward energy transferred per collision:

$$\Delta E_d = \Delta E_{d300} \left(\frac{T}{300 \text{ K}} \right)^n \text{ kJ/mol}$$

where $\Delta E_{d300} = 3.47 \text{ kJ/mol}$ (for CH_2OCHO , OCH_2OCHO , HOCH_2O , and $\text{CH}_3\text{OCH}_2\text{O}$) is adopted from a previous study of vinyl + O_2 [79], and the temperature exponent, $n = 0.7$, is adopted from previous studies for N_2 as the bath gas [80]. Lennard-Jones (LJ) collision diameters and well depths were estimated via the Joback method [81], which is based on the critical temperature and pressure of parent compounds. The method of corresponding states was subsequently used to estimate the LJ parameters [82] and is described in the context of RMG elsewhere [83].

3.2. Experiment simulation details

The Python 3.4 interface of Cantera [84] version 2.2.0 was used for all simulations in this paper other than premixed laminar flame speed calculations, for which CHEMKIN-PRO [85] was used. In Cantera, state information about the system is stored in an instance of the Solution class. The Solution classes used in this study model simple, compressible systems and, as such, require two independent intensive variables to fix the state. These instances of Solution classes are further utilized to compute the thermodynamic properties required to solve the ODEs representing the combustion systems studied here.

Cantera [84] contains several objects used to model homogeneous reacting systems; the two used in this paper are a Reservoir and an IdealGasReactor, which are subclasses of the generic Reactor class. The specific IdealGasReactor class is preferred over the generic Reactor class in this study because the energy equation is directly solved in terms of the temperature, instead of the enthalpy. A Solution object is installed in each Reactor subclass to manage the state information. The difference between the

IdealGasReactor and the Reservoir is simply that the state (i.e., the pressure, temperature, and chemical composition) of the Solution in a Reservoir is fixed.

3.2.1. Shock tube simulations

Comparisons with experiments were made by assuming all shock tube ignition delay time measurements could be modeled as 0-D constant volume, adiabatic, batch reactors (i.e., constant internal energy and volume reactors). This assumption is valid for experiments with negligible rates of change of pressure and temperature after the shock wave reflects off the tube endwall prior to the ignition event. We note, however, that traditional shock tube ignition delay measurements may show boundary layer, mixture attenuation, and non-homogeneous effects that can reduce the observed τ compared to a corresponding ideal scenario for long ignition delay times [86]. These conditions cannot be reliably modeled with the constant internal energy-volume (CONV) approach. This behavior and best practices for minimizing its influence in both experiments and models have been discussed in great detail elsewhere [86,87]. We therefore recognize the possibility of reduced reliability in model-data comparisons for shock tube targets with a long τ .

Unless otherwise noted, all shock tube ignition delay times were defined as the time to reach the simulated maximum slope of $[\text{OH}]$ with respect to time after the arrival of the shock wave at the endwall. Simulations of species profiles measured behind reflected shock waves in DME/argon mixtures at the Stanford High Temperature Gasdynamics Laboratory [45] were conducted in the same manner as the τ simulations.

Sensitivity analyses for CONV simulations were performed in Cantera [84] and with respect to $[\text{OH}]$. In this paper, normalized sensitivity coefficients of $[\text{OH}]$ with respect to individual model rate parameters, k_i , were computed:

$$S_{[\text{OH}],k_i} = \frac{\partial \ln [\text{OH}]}{\partial \ln k_i}$$

For all analyses, a user-specified time was chosen (e.g., time of $\max(d[\text{OH}]/dt)$, time at 50% fuel consumption, etc.). Sensitivity parameters were then further normalized to -1 or 1 based on the largest value by magnitude within the sensitivity array.

3.2.2. Rapid compression machine simulations

As mentioned previously, non-reactive experiments were conducted to characterize the machine-specific effect on the experiments in the RCM. A non-reactive experiment was conducted for each reported ignition delay in this study. The non-reactive pressure trace was combined with the reactive pressure trace to produce a combined pressure trace. For this, the reactive pressure trace was taken until the EOC, and the non-reactive pressure trace was taken after the EOC.

The combined pressure trace was used to compute a volume trace that could be applied to simulations to include any facility effects. The volume trace was created by initializing a Cantera Solution object at the initial pressure and temperature (P_0 , T_0) of the reactor. After initialization, the initial mass-specific entropy (s_0) and density (ρ_0) were recorded. The initial volume (V_0) was arbitrarily taken to be equal to 1.0 m^3 ; the initial volume in this step was arbitrary so long as the initial volume utilized in the Reactor-based simulations described below was the same, since the velocity (i.e., the time rate of change of volume) enters the energy equation, not the volume itself.

The measured pressure at each data point (P_i) from the combined pressure trace was used with the previously recorded initial entropy (s_0) to set the state of the Solution object sequentially. At each data point, the volume was computed according to

conservation of mass and the ideal gas law

$$V_i = V_0 \frac{\rho_0}{\rho_i}$$

where ρ_i was the density at each point computed by the Cantera Solution. This procedure effected an isentropic compression/expansion process, where compression occurred during the compression stroke and expansion occurred after the EOC. Note that the expansion process after the EOC was not a physical process; it was simply a modeling of the inherent experimental heat loss post-EOC. Moreover, this procedure accounted for the variation of the specific heat ratio as a function of temperature. Since there were no significant reactions that occurred during the compression stroke for the reactive experiments, or during and after the compression stroke for the non-reactive experiments, it was not necessary to solve the species and energy equations using an IdealGasReactor. After generation, the volume trace was downsampled to 20 kHz (from the experimental rate of 100 kHz) to reduce computational time of the simulations. This has no effect on the simulations, as confirmed by applying the full-rate (i.e., 100 kHz) volume trace to a simulation.

This volume trace was applied to a reactive simulation in Cantera [84] as follows. A Solution object was initialized to the initial state of the reactor (P_0 , T_0) and installed in an IdealGasReactor object ($V_0 = 1.0 \text{ m}^3$). A Reservoir object was defined containing an arbitrary state, and a Wall was installed between the Reservoir and the Reactor. The Wall allows the volume of the IdealGasReactor to vary with time, and the effect of the Wall enters the energy equation through the velocity term as the Wall does work on the IdealGasReactor. The velocity of the Wall was specified based on the first forward difference of the volume as a function of time.

Finally, the IdealGasReactor was installed in a ReactorNet, which implements the interface to the solver CVODES. CVODES is an adaptive-time-stepping solver, distributed as part of the SUNDIALS suite [88]. As the solver stepped towards the end time of the simulation, the state of the system was stored on each integrator time step, producing simulated pressure, volume, and temperature traces. Comparisons between the simulated pressure traces from Cantera [84] and CHEMKIN-PRO [85] show negligible differences.

Simulations were conducted for two purposes – to determine the temperature at the EOC, T_c , and to compute the simulated ignition delay. The temperature at the EOC was used as the reference temperature for this RCM data and was therefore reported for all of the data points collected in this paper. T_c was defined as the simulated temperature at the EOC. To ensure that the adiabatic assumption during the compression stroke was realized, the simulated pressure during the compression stroke for a reactive simulation and a simulation where reactions have been “turned off” (e.g., by setting the reaction rate multiplier to zero) were compared and found to be identical. In addition, the non-reactive simulated pressure trace was compared to the non-reactive experimental pressure trace and no differences were found.

Two definitions of the ignition delay in the simulations were used. The first definition matches the definition in the experiments, with the overall ignition time defined as the global maximum of the time derivative of the pressure (dP/dt) after the EOC, and the first-stage ignition delay time defined as the first maximum of dP/dt after the EOC. The time derivative of the pressure in the simulations was computed by second order Lagrange polynomials, as discussed by Chapra and Canale [89]. In some cases, however, the pressure rise during the first-stage ignition was slow, resulting in a small peak in dP/dt . Due to electrical noise in the measured pressure signal that propagated into the simulations through the computed volume trace, a first peak in dP/dt was not always observed in the simulations despite a significant deviation of the simulated reactive pressure trace from the non-reactive

pressure trace prior to ignition. In such cases, the first peak of the overall simulated heat release rate (HRR) was used to define a first-stage ignition delay time, $\tau_{1,\text{HRR}}$. This definition of the first-stage ignition delay was found to align well with the definition of first-stage ignition delay by the first maximum of dP/dt . A comparison of these definitions is shown in Fig. 1b. Note that Fig. 1a and b show different T_c conditions; this is because we could not find an overlapping condition where the experimental dP/dt , the simulated dP/dt , and the HRR all showed distinct peaks during the first-stage ignition event.

3.2.3. Jet-stirred reactor simulations

Jet-stirred reactors were modeled as 0-D adiabatic, constant pressure, perfectly-mixed, isothermal reactors with no reactions at the walls. A Cantera Solution object was installed in an IdealGasReactor. Subsequently, the IdealGasReactor was connected to a Cantera Valve object with inlet flow from a Reservoir supplying the reactant mixture. The IdealGasReactor was also connected to an outlet – a Cantera MassFlowController object, whose mass flow rate was calculated from a corresponding experimental residence time, with exhaust products flowing into another Cantera Reservoir object. This network of objects (Inlet→IdealGasReactor→Outlet) was installed in an instance of ReactorNet and integrated until steady state was reached. In all cases, integrating from multiples of 10–25 in residence times was sufficient for the systems to reach steady state.

3.3.4. Premixed laminar counterflow and burner-stabilized flame simulations

Simulations were performed using CHEMKIN-PRO [85]. All calculations in this paper were performed with the multicomponent transport formulation and inclusion of thermal diffusion. Domain-discretized grid sizes of 200 or more points were obtained for each simulation, which was deemed sufficiently resolved (as demonstrated in a benchmark exercise for simulations of up to 600 grid points). Similarly, burner-stabilized premixed laminar flames were modeled with imposed experimentally measured temperature profile constraints. Multicomponent transport and thermal diffusion were included. Inlet unburned gas velocities were used as a boundary condition for the flame model.

3.3. RMG simulations

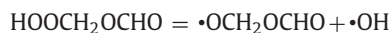
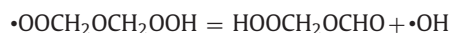
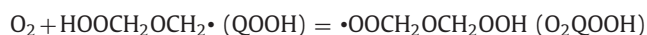
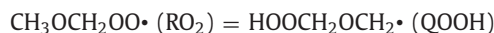
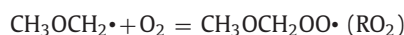
The open source Reaction Mechanism Generator (RMG) [31], Python version 1.0 (found at <http://reactionmechanismgenerator.github.io>) was used for automated kinetic model construction. Details of RMG's algorithm and methods can be found elsewhere [90] and are described briefly here. The model is grown using the rate-based algorithm developed by Susnow et al. [91], where RMG simulates isothermal, isobaric reactor conditions with a user-given set of initial “core” species, using a set of reaction family templates to discover all possible reactions between them, forming a new set of “edge” species. At each integration step, RMG evaluates whether an edge species is important enough to be included in the model by checking whether it has a larger flux than the user-specified tolerance. If so, the edge species is added to the set of core species constituting the model, and it is reacted with all the existing core species to generate additional edge species and reactions. The simulation is repeated from $t = 0$ until RMG deems another edge species to be important. This iterative process of adding new edge species and re-simulating the system continues until the model satisfies the user-given termination criteria and all fluxes to edge species are below the threshold tolerance. For more information on this software package, see Gao et al. [92].

4. Results and discussion

4.1. Detailed kinetic models

4.1.1. Base model

A base model (hereafter described as Model I) of 120 species and 701 reactions is assembled in this paper from three principal sources. The propane submechanism (along with the majority of the small molecule chemistry) is adopted from Merchant et al. [28], in which the low-temperature kinetics are derived from the detailed *ab initio* and rate theory calculations of Goldsmith et al. [27]. Goldsmith and coworkers also computed the thermodynamic parameters for a large number of the species relevant to propane oxidation [27]; these are adopted in this paper. The base chemistry for $\text{H}_2/\text{CO}/\text{H}_2\text{O}_2/\text{HCO}$ compounds was updated with that from the CEFRC foundational fuel mechanism [93,94]. The remaining base chemistry in the model assembled by Merchant et al. [28] is derived from the work of Healy et al. [95] and is carried forward into Model I. The DME kinetics and related thermochemistry are largely adopted from the recent work of Burke et al. [9], who have validated their model for a number of experimental targets over a wide range of pressure, temperature, and composition conditions. Those authors also computed pressure-dependent rate parameters for the decomposition of CH_3OCH_2 and pathways relevant to the first O_2 addition. Between 650 and 1000 K, the $\text{CH}_3\text{OCH}_2\text{O}_2$ adduct formed from the first O_2 addition to CH_3OCH_2 is consumed primarily by the well-known isomerization-elimination chain branching pathway:



This first O_2 addition and its subsequent isomerizations and decomposition reactions have also been recently investigated by Eskola et al. [16] and Rodriguez et al. [19]. We tested the use of all three (the two aforementioned sets in addition to that by Burke et al. [9]) elementary reaction sets against a wide range of low-temperature experimental targets, finding that the reaction sequence by Burke et al. [9] gave superior overall performance against the large number experimental targets used in this paper. Thus, we retained the reactions corresponding to the first O_2 addition from the model of Burke et al. [9]. The rate of $\text{CH}_2\text{O} + \text{H} = \text{HCO} + \text{H}_2$ was also updated based on the results of a recent combined computational and theoretical study [96]. Further updates based on rate evaluations and calculations are described below.

4.1.2. RMG model

One additional model was generated in this paper using RMG. By specifying Model I as a “seed mechanism” in RMG, all of the reactions in Model I were forced to be included in the final RMG model. This model is referred to hereafter as Model II. Reactor conditions spanning the temperature and pressure ranges of the RCM data obtained in this paper were used as RMG input reactor parameters to generate Model II. This model is substantially larger in size, with 168 species and 3979 reactions, as RMG found more than 3200 reactions and 48 species that satisfied the criteria for

Table 3

Species and reaction count for models used in this work.

Model	Number of species	Number of reactions
Model I	120	701
Model II*	168	3979

* RMG-generated species are listed in the Supplementary material.

inclusion in the model. Aspects of the two models generated in this paper are summarized in Table 3.

Model II was used to simulate many of the experimental targets chosen for this paper, including the RCM data measured in the present study and the JSR data of Rodriguez et al. [19]. For all cases considered, Model I and II predicted the available experimental data equally well, even though RMG discovered a plethora of reactions between DME and propane radical intermediates. Although it is possible that differences between Models I and II may surface for fuel-rich or pyrolytic conditions, these are not conditions of practical interest since most practical devices [ideally] operate with fuel-lean or stoichiometric combustion. We note, however, that Model II did have an impact on predicted methyl formate JSR species profiles of Rodriguez et al. [19], which are sensitive to a number of factors discussed below. Model I is used as the primary comparison tool with the experimental data in the remainder of this paper. In addition to the model-experiment comparisons presented in this paper, a large number of additional validations using Model I are available in the Supplementary material and are identified in Table 1.

The results of this exercise support the hypothesis that binary fuel models designed for dual fuel purposes may be assembled simply based on combinations of their separate submechanisms. This notion, in combination with advanced model merging tools [97] which can identify and eliminate redundant species and/or reactions, may make model generation for future binary fuel mixtures a facile task (assuming the independent submechanisms are reliable). Although cross-reactions between larger radical intermediate derivatives of DME and propane do not influence any of the experimental targets modeled in this paper, there is clearly kinetic coupling of the fuels as a result of DME's higher reactivity and its OH, HO_2 radical formation which promotes autoignition of propane. As illustrated and elaborated upon in a later section, even small amounts of DME can significantly reduce the ignition delay times of DME/propane binary mixtures.

4.2. Rate considerations, evaluations, and calculations

As discussed in the previous section, a number of updates to rates of important reactions in the DME and propane submechanisms were performed in this paper. These are described in the following sections. Regarding rates computed in this paper, CCSD T1 diagnostic values (tabulated in the Supplementary material) for almost all of the transition states exceed 0.02, suggesting multi-reference wavefunction character. Thus, the computed barrier heights are likely uncertain by over 1 kcal/mol. Adding to the uncertainty in computed global rates are uncertainties in the solution to the master equation for energy transfer and resulting kinetic pressure-dependence. A rigorous propagation of uncertainties to overall rate uncertainty parameters is beyond the scope of this paper, albeit we estimate that the total high-pressure limit rates are accurate to within a factor of three and rates in the falloff regime uncertain by up to a factor of five.

4.2.1. $\text{HO}_2\text{CH}_2\text{OCHO} = \text{OH} + \text{OCH}_2\text{OCHO}$

The decomposition of $\text{HO}_2\text{CH}_2\text{OCHO}$ (hydroperoxymethylformate) to $\text{OH} + \text{OCH}_2\text{OCHO}$ is known to drive autoignition of DME

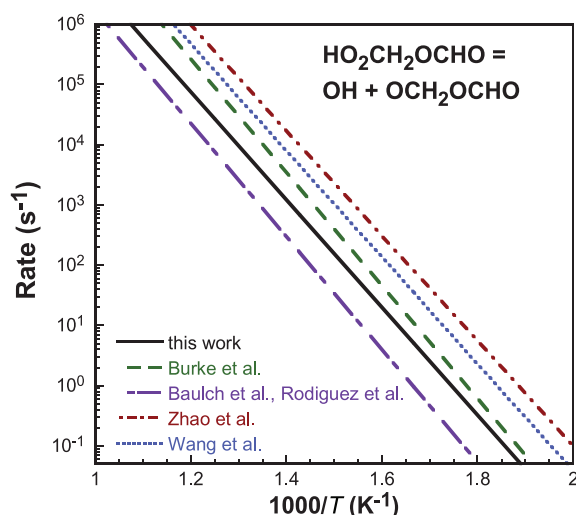
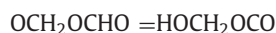


Fig. 2. Comparison of the rates of the decomposition of the ketohydroperoxide hydroperoxymethylformate ($\text{HO}_2\text{CH}_2\text{OCHO}$) to $\text{OH} + \text{OCH}_2\text{OCHO}$ between 500 and 1000 K. Black line: this work; red dash-dotted line: Zhao et al. [20]; blue dotted line: Wang et al. [42]; green dashed line: Burke et al. [9]; purple long/short-dashed line: Baulch et al. [98], for the reaction $\text{C}_2\text{H}_5\text{OOH} = \text{C}_2\text{H}_5\text{O} + \text{OH}$, recommended between 400–800 K (used in Rodriguez et al. [19]). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

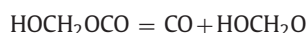
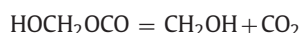
below 700 K [20]. The rate of this decomposition reaction remains highly uncertain and is typically tuned to fit experimental data (published and adopted rates can differ by over two orders of magnitude). In addition, all current DME models use a rate derived from the soft recommendation of Baulch et al. [98] for the decomposition of $\text{C}_2\text{H}_5\text{OOH}$ (ethyl hydroperoxide) to $\text{C}_2\text{H}_5\text{O} + \text{OH}$ (although most others have adjusted the rate coefficient). The decomposition rate of the ketohydroperoxide is not expected to be the same as the ethyl hydroperoxide. A similar approach was adopted in this paper, where the activation energy of the nominal Baulch evaluated rate [98] was reduced by 2 kcal/mol to better match the DME RCM experiments conducted in this work. Figure 2 compares the rate of this decomposition reaction with select literature values, illustrating the large disparity. As elaborated upon below, the assumed rate of this reaction also strongly influences predicted peak methyl formate (MF) mole fractions in the JSR experiments of Rodriguez et al. [19].

4.2.2. $\text{OCH}_2\text{OCHO} = \text{HOCHO} + \text{HCO}$

Prior to a recently discovered concerted pathway, the resulting OCH_2OCHO radical was thought to solely undergo an intra-H migration to HOCH_2OCO :



Under the low-temperature oxidative conditions where HOCH_2OCO is important in DME combustion, it can undergo further unimolecular decomposition:



where the α -hydroxymethyl (CH_2OH) radical will almost exclusively react with O_2 under the conditions where the above pathway is important, although at higher temperatures it may undergo H-elimination to $\text{CH}_2\text{O} + \text{H}$ [99]. HOCH_2O may undergo a variety of decomposition reactions:

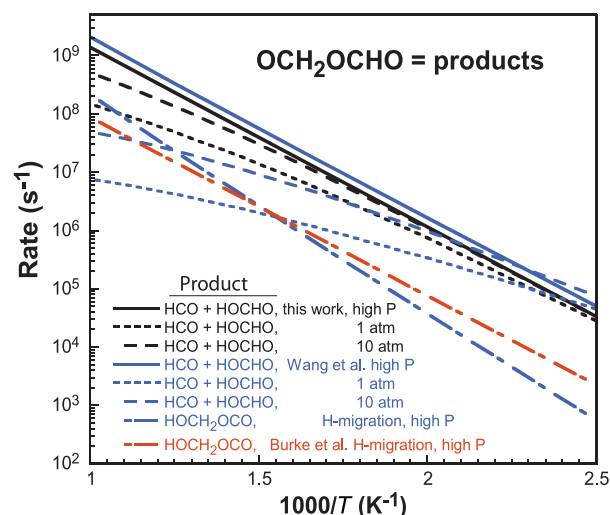
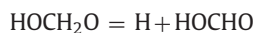
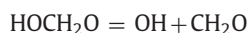
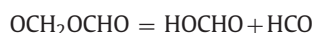


Fig. 3. Arrhenius comparison of rate constants at various pressures for the unimolecular decomposition of OCH_2OCHO . Black: this work; blue: Wang et al. [42]; orange: isomerization to HCH_2OCO (concerted elimination to $\text{HCO} + \text{HOCHO}$ was not considered) [20] [9]. In the work of Rodriguez et al. [19], different products are assumed. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



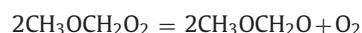
which are further discussed below in a separate section. Wang et al. [42] recently discovered a concerted elimination from OCH_2OCHO :



This concerted reaction results in formic acid (HOCHO) and formyl (HCO), and is faster than the competing intra-H migration by an order of magnitude or more below 1400 K. In their work, Wang et al. [42] explored the potential energy surface in detail at the UQCISD(T)/CBS//B3LYP/6-311++G(d,p) level of theory, followed by pressure-dependent rate calculations using MESMER [100]. Rate coefficients were also computed in this paper, as described above, and a comparison of the decomposition with other published values is shown in Fig. 3. As illustrated in Fig. 3, there is considerable pressure-dependence at higher temperatures. Although the high-pressure limit rates between this paper and Wang et al. [42] are in good agreement, there exists notable disagreement between the 1 atm and 10 atm rate constants. In their work, Wang et al. [42] used RRHO for all internal degrees of freedom, which could explain the slightly higher high-pressure limit rate in comparison to that computed here. Regarding pressure-dependence, our computed rate coefficients differ from those computed by Wang et al. [42] by about an order of magnitude in the combustion-relevant temperature ranges. In both the RRKM/ME simulations conducted here and in the work of Wang et al. [42], the exponential down model is used for describing collisional energy transfer. At 1000 K, the average downward energy transferred using the parameters in this paper is 2.5 times greater than that used in the model of Wang et al. [42], which was chosen based on previous work by Carr et al. [101] in a study of the pressure-independent reaction of $\text{DME} + \text{OH}$. As stated in the work of Carr et al. [101], the parameters of the energy transfer model were arbitrarily chosen, and as further explained by the authors, they do not have an effect on the computed results since the reaction under study is

not pressure-dependent to begin with. This is in contrast to the energy transfer parameters used here, which were not arbitrarily chosen. Thus, the rates computed here are recommended over those computed by Wang et al. [42]. It is also noted that in the recent work of Rodriguez et al. [19], the reaction products were assumed to be $\text{OCH}_2\text{OCHO} = \text{CH}_2\text{O} + \text{OCHO}$, negating the need to include the species HOCH_2OCO in their reaction mechanism (along with its subsequent reactions). The recent mechanism of Burke et al. [9] included estimates for the intra-H migration channel, as well as a decomposition pathway to $\text{CH}_2\text{O} + \text{OCHO}$.

At low temperatures, the self-reaction of $\text{CH}_3\text{OCH}_2\text{O}_2$ (methoxymethyl peroxy, the 'RO₂' corresponding to DME) is in competition with $\text{RO}_2 = \text{QOOH}$ isomerization and subsequent OH producing reactions. Two possible product channels are considered:



Determination of product yields of this reaction has been hampered by the fact that both reactions yield MF, although Jenkin et al. [102] reported a 0.7 ± 0.1 branching ratio from FTIR product studies and photolysis experiments involving methoxymethyl peroxy. Thus, the branching fraction is dominated by the first pathway. This is further discussed below.

The rate of the RO_2 self-reaction was also indirectly measured by Rosado-Reyes et al. [103] in the temperature range of 295–600 K and pressures of 20–200 torr. In their study, the authors investigated branching ratios of the $\text{CH}_3\text{OCH}_2 + \text{O}_2$ reaction by monitoring yields of formaldehyde (CH_2O), methyl formate, and formic acid via transient infrared spectroscopy. The rate constant of the methoxymethyl peroxy radical self-reaction is calculated from the kinetics of the formaldehyde and methyl formate product yields [103]. At 298 K, the resulting rate of $1.9 \times 10^{12} \text{ cm}^3/\text{mol}\cdot\text{s}$ is in good agreement with that measured by Jenkin et al. [102] at the same temperature ($1.3 \times 10^{12} \text{ cm}^3/\text{mol}\cdot\text{s}$). The mechanism of Zhao et al. [20] used a combination of the results from Dagaut et al. [104] and Jenkin et al. [102], originally implemented by Fisher et al. [17] and Curran et al. [18]. Both Dagaut et al. [104] and Jenkin et al. [102] noted that this reaction is pressure-dependent and in the falloff regime between 228–380 K and 25–800 torr. Although Dagaut et al. [104] provided a pressure-dependent parameterization of the self-reaction by monitoring its decay over 228–380 K and 25–800 torr, no kinetic model incorporates it. Furthermore, the reported experimental uncertainties of Dagaut et al. [104] are smaller than what has been reported by Rosado-Reyes et al. [103]; thus, the tabulated rates of Dagaut et al. [104] were parameterized into the PLOG format and used in this paper. Additionally, the branching fraction was assigned a value of 0.7 (in favor of the $2\text{CH}_3\text{OCH}_2\text{O} + \text{O}_2$ channel), as observed by Jenkin et al. [102]. A comparison of the resulting rate used in this paper is shown with other values in Fig. 4. Up to 600 K, the rates adopted in this paper fall within the lower bound of the self-reaction rate constant reported by Rosado-Reyes et al. [103], also illustrated in Fig. 4.

4.2.3. $\text{CH}_3\text{OCH}_2\text{O} = \text{CH}_3\text{OCHO} + \text{H}$

This alkoxy species is predominantly formed during the self-reaction of methoxymethyl peroxy radicals, as described above. Upon its formation, methoxymethyloxyl very quickly decomposes into methyl formate (CH_3OCHO) + H (see Fig. 5 and illustration of computing $\text{CH}_3\text{O} + \text{CH}_2\text{O}$ product channel from [20]). Although rate calculations for this reaction have been performed by Song et al. [105], no rate expressions were given. Thus, the temperature- and pressure-dependent kinetics were computed in this paper.

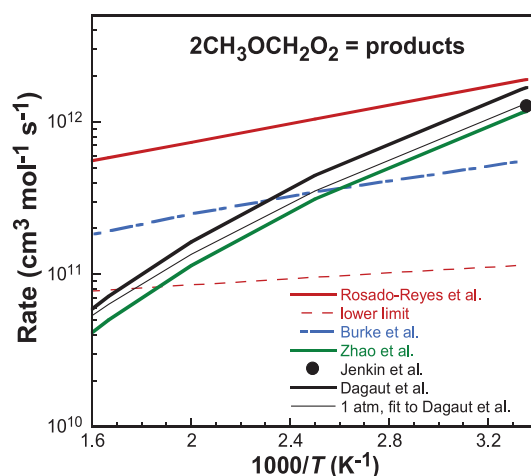


Fig. 4. Arrhenius plot of the methoxymethyl peroxy self-reactions dominant channel plotted between 298 and 600 K. Black line: high-pressure limit rate from Dagaut et al. [104]; thin black line: fitted 1 atm rate from Dagaut et al. [104]; red: Rosado-Reyes et al. [103] (with lower limit as thin red line); green line: Zhao et al. [20], which is a combination of the results from Dagaut et al. [104] and Jenkin et al. [102], originally implemented by Fisher et al. [17] and Curran et al. [18]; black filled circle: experimental result of Jenkin et al. [102]; blue short/long dashed line: Burke et al. [9]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

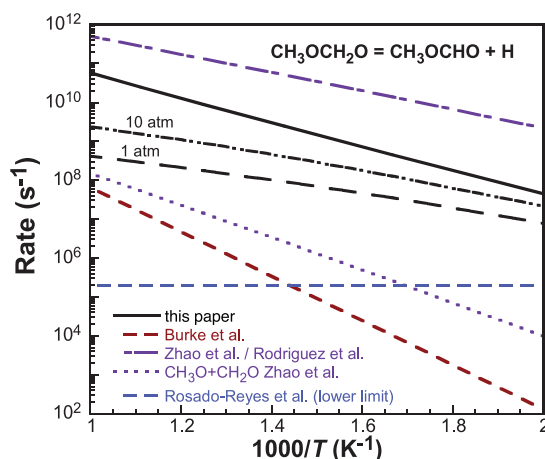


Fig. 5. Comparison of rate constants for the decomposition of $\text{CH}_3\text{OCH}_2\text{O}$ to methyl formate (CH_3OCHO , MF) + H. Black: this paper, with 1 atm and 10 atm rates also included; red: Burke et al. [9]; blue: Rosado-Reyes et al. [103], lower bound; purple: Zhao et al. [20] (adopted by Rodriguez et al. [19]) with short dashed purple lines representing the decomposition channel to $\text{CH}_3\text{O} + \text{CH}_2\text{O}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In the work of Rosado-Reyes et al. [103], the alkoxy decomposition rate is assigned a lower bound of $2 \times 10^5 \text{ s}^{-1}$, as depicted in Fig. 5. This value was fitted to MF and CH_2O profiles at 503 K and defined as a lower limit since the rate-limiting reaction step under the authors' experimental conditions is the self-reaction of the methoxymethyl peroxy radicals ($\text{CH}_3\text{OCH}_2\text{O}_2$). At 500 K, the rate of the corresponding high-pressure limit reaction computed in this paper is over two orders of magnitude faster. Compared to the rate computed here, large discrepancies in rates exist with those used in other models. As seen in Fig. 5, the rate computed here roughly splits the values determined/used in previous work. The decomposition rate is very fast: at 500 K and 1 atm, the lifetime of $\text{CH}_3\text{OCH}_2\text{O}$ is less than 0.1 ms. Thus the overall self-reaction of methoxymethyl peroxy can be represented as:

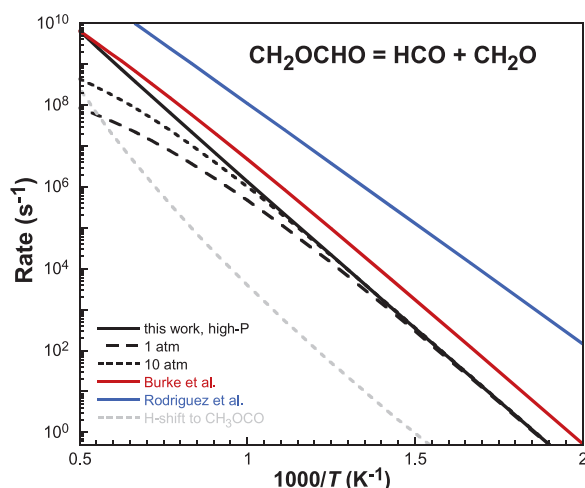


Fig. 6. Comparison of rate constants at various pressures for the decomposition of CH_2OCHO . Black: this work; red: Burke et al. [9]; blue: Rodriguez et al. [19]; light grey: this work, intra-H migration to CH_3OCO . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

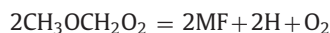


Figure 5 also shows the minor channel ($\text{CH}_3\text{OCH}_2\text{O} = \text{CH}_3\text{O} + \text{CH}_2\text{O}$) from the model of Zhao et al. [20]; the rate of reaction is not competitive with the MF channel, and the reaction was therefore removed from the final model.

4.2.4. $\text{CH}_2\text{OCHO} = \text{CH}_2\text{O} + \text{HCO}$

This radical species is a derivative of methyl formate, which is formed during the oxidation of DME. This reaction is only included at the high-pressure limit in current combustion models [9,19,106,107]. Thus, the pressure-dependence was computed here. At higher temperatures, the pressure-dependence of this reaction may influence the combustion kinetics of methyl formate via H-abstraction reactions by various radicals from the parent methyl formate, forming CH_2OCHO . The decomposition rate is compared against other literature values in Fig. 6. The high-pressure limit used in the model by Burke et al. [9] is in fair agreement with that computed here, although that used by Rodriguez et al. [19] is faster by 1–3 orders of magnitude from 298–2000 K. Although intra-H migration is possible for this system, the unimolecular decomposition dominates for all temperatures, as illustrated in Fig. 6.

4.2.5. $\text{HOCH}_2\text{O} = \text{H} + \text{HOCHO}$

HOCH_2O is a radical derivative of methanediol and can be formed during its oxidation. In the context of DME, all combustion kinetic models for its oxidation created before the year 2015 (before inclusion of the concerted elimination of OCH_2OCHO to $\text{HOCHO} + \text{HCO}$ computed by Wang et al. [42] and in this paper) predict the formation of large quantities of HOCH_2O during low-temperature DME oxidation simulations. However, because the intra-H migration from OCH_2OCHO to HOCH_2OCO is in fact a minor channel, HOCH_2O forms only in small amounts during DME oxidation. Because its rate of decomposition is expected to be pressure-dependent, and because this alkoxy radical may be an important intermediate during the oxidation/pyrolysis of other potential alternative fuels, its pressure-dependent decomposition rates were computed here. An Arrhenius plot comparing the results of the RRKM/ME simulations conducted in this paper are shown in Fig. 7 with comparisons to other available rates. We note that $\text{HOCH}_2\text{O} = \text{OH} + \text{CH}_2\text{O}$ is also included in the detailed kinetic model (written in the reverse direction, and adopted from Xu et al.

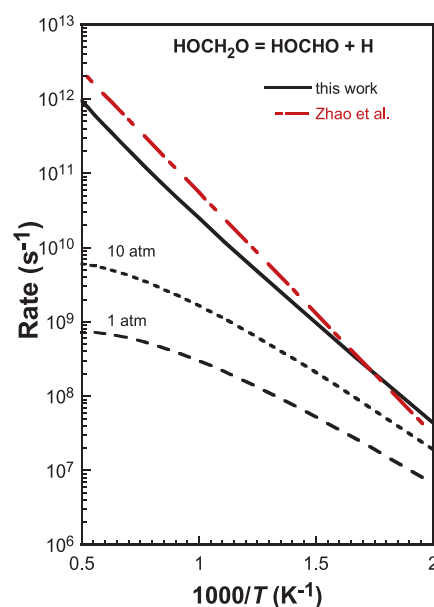
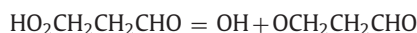


Fig. 7. Arrhenius plot for the decomposition of HOCH_2O . Black: this work; red: used in a number of other models [20,9,42]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

[108]), but this reaction is slower under all temperatures (three orders of magnitude at 600 K, increasing to 5% of the $\text{H} + \text{HOCHO}$ channel at 1500 K).

4.2.6. Role of cyclic peroxides

Ketohydroperoxides (KHPs) produced during low-temperature hydrocarbon oxidation may undergo Korcek decomposition reactions [109–112] that can directly compete with decomposition to OH and an alkoxy radical. The Korcek decomposition reaction of a DME KHP (hydroperoxymethyl formate) is to form two formic acids or carbonic acid + formaldehyde. Carbonic acid has been recently detected by mass spectrometry after molecular beam sampling from a JSR by Moshhammer et al. [113]. However, no rate calculations have been performed for this reaction, because the Korcek reaction for $\text{HO}_2\text{CH}_2\text{OCHO}$ is not expected to play a significant role under the conditions of interest here (see the comparison for a propane KHP below). The primary source of OH radicals during the low-temperature oxidation of propane is through the $\text{HO}_2\text{CH}_2\text{CH}_2\text{CHO}$ KHP and its unimolecular decomposition [114]:



which is in competition with formation of a cyclic peroxide (CP):



The CP may further react through the following reactions:



These pathways are also schematically illustrated in Fig. 8. The pathway to a CP and its subsequent products was considered in this paper, but inclusion of the rates recently computed by Jalan et al. [114] did not have any influence on simulation results. The reason for this is evidenced by a comparison of the KHP decomposition rates, illustrated in Fig. 9. Under the lowest temperature RCM experiments of this paper (~ 700 K), decomposition of KHP to $\text{OH} + \text{OCH}_2\text{CH}_2\text{CHO}$ dominates by three orders of magnitude. Only

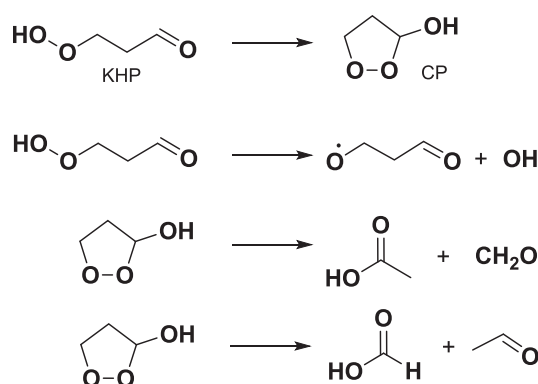


Fig. 8. Ketohydroperoxide (KHP) fate and cyclic peroxide (CP) decomposition reactions investigated in previous works, with respect to propane oxidation [114,27].

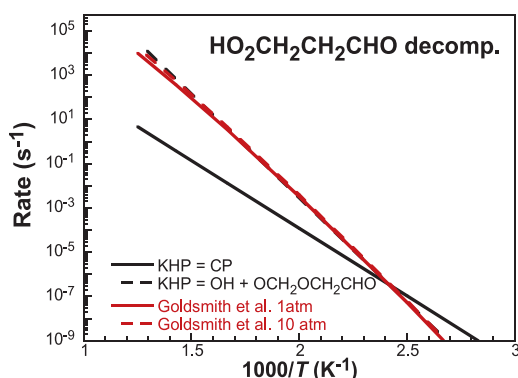


Fig. 9. Rate comparison of KHP (HO₂CH₂CH₂CHO) decomposition pathways. Black: high-pressure limit rates from Jalan et al. [114]; Red: Goldsmith et al. [27]; see Fig. 8 for species definitions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

at and below 400 K does CP formation become competitive with the aforementioned channel. Thus, formation of CP and its subsequent reactions is not included in the kinetic mechanism in this paper.

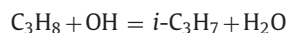
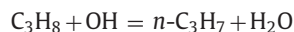
Last, it must be noted that there also exists the possibility of a formic acid catalyzed conversion of KHP to CP [114], which is relevant to the low-temperature JSR data of Rodriguez et al. [19]. Jalan et al. [114] used high-level quantum chemical methods for the analogous propane mechanism, predicting that the formic acid catalyzed reaction can lower the barrier by over 30 kcal/mol. To test the influence of this pathway on the simulated species profiles for the JSR data investigated here, rate parameters for the acid catalyzed isomerization and subsequent decomposition were estimated by analogy to those for propane (in this case, the resulting DME CP is assumed to decompose into two formic acid molecules). JSR simulations were then performed with these two additional reactions:



Simulations of stoichiometric DME/O₂ mixtures in helium (corresponding to the conditions in the work of [114]) resulted in accumulation of CP up to 100 ppm (10% of the KHP peak concentration), peaking at the same temperatures as HO₂CH₂OCHO. In addition, peak formic acid and HO₂CH₂OCHO concentrations were predicted to change by ~10%. Thus, future work should include exploration of this acid-catalyzed mechanism if the low-temperature ($T < 600$ K) JSR product distributions are to be accurately predicted.

4.2.7. C₃H₈ + OH

The competition between primary and secondary H-abstraction by OH on propane is critically important in dictating its autoignition behavior. Two reactions are possible:



As noted by Merchant et al. [28], H-abstraction from the methyl groups of propane promotes low-temperature autoignition due to the resulting OH chain branching pathways, while H-abstraction at the secondary carbon site retards ignition because it leads to HO₂ formation, which further undergoes HO₂–HO₂ recombination. Therefore, we have derived rates that accurately capture both the branching ratios and total rate constants over a wide temperature range. Droege and Tully [115] previously measured absolute rate coefficients and branching fractions between 293 and 854 K through a laser photolysis/laser-induced fluorescence technique at 400 torr (in helium) with branching ratios determined by performing experiments using six selectively deuterated isotopes of propane. The resulting branching ratios were found to be strikingly similar to those previously determined by Walker's fits to experiments at 297 K and 753 K [116]. More recently, rates and branching ratios of this reaction have been computed by Huynh et al. [117], who compared their predictions with numerous other efforts performed over the last 30 years, finding excellent agreement with the experimental work of Droege and Tully [115]. However, the authors did not publish their computed rates in any form other than in an Arrhenius plot. Thus, the branching ratios published by Droege and Tully [115] were fitted to functional forms in this paper – a power law form was used for the *i*-propyl branch, while a log form was used for the *n*-propyl branch with fitting errors less than 5% below 1500 K. These were then used to define the rates for each channel based on the total rate fitted by Sivaramakrishnan et al. [118], which accurately captures experimental data spanning a wide temperature range. Sivaramakrishnan et al. [118] measured the total rate of C₃H₈ + OH in the temperature range of 797–1259 K. In their study, they performed a modified three-parameter Arrhenius regression of their data along with those of 13 other studies spanning the temperature range of 190–1248 K. This total rate – in combination with the branching fits from the work of Droege and Tully [115] – was used to derive new C₃H₈ + OH rate constants. Last, we note that the total rate derived here also agrees within 10% of a recent determination by Badra et al. [119] for temperatures between 600 and 2000 K. Because their experiments were performed near 1000 K and the focus of this paper is on low-temperature oxidation, the branching ratios measured by Droege and Tully [115] between 293 and 854 K were retained in this paper. Nevertheless, site specific rate constants for between the two aforementioned studies are in near perfect agreement below 1000 K.

Comparisons with this newly derived rate are made with other available rates in Fig. 10. The new rates were adopted in this paper and also fitted to a modified three-parameter Arrhenius form and are given in Table 4.

4.3. Low-temperature propane oxidation

RCM ignition delay measurements were obtained for pure propane/oxidizer mixtures in order to increase model fidelity for low-temperature propane autoignition and at engine-relevant conditions. Experiments were performed for three different equivalence ratios in the range $T = 690$ – 910 K. Comparisons of this data with the model of this paper is shown in Fig. 11. A normalized sensitivity analysis of rate parameters to [OH] at τ to the model

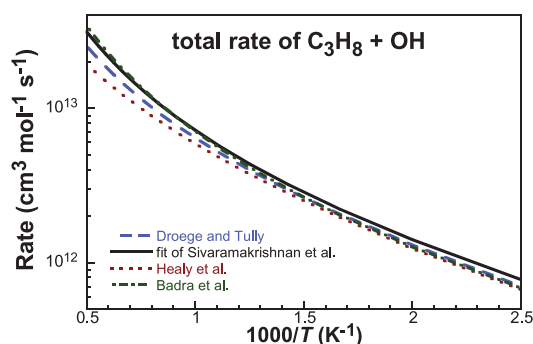


Fig. 10. Comparison of the total rate of H-abstraction by OH on propane. Black solid line: fit of Sivaramakrishnan et al. [118], used in this work; blue dashed line: Droege and Tully [115], as well as DeSain et al. [120,121]; red dotted line: Healy et al. [95]; green dashed-dotted line: Badra et al. [119]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Three-parameter modified Arrhenius fits to reaction rates studied in this work, computed at the CCSD(T)-F12a/cc-pVTZ-F12//M08SO/MG3S level or re-evaluated. Units in cm^3 , mole, cal, and s. Pressure-dependent rate parameterizations included in the Supplementary material.

Reaction channel	High-pressure limit rate parameters ^a		
	A	n	E_a (cal/mol)
$\text{HOCH}_2\text{O} = \text{HOCHO} + \text{H}$	4.83E+08	1.350	10,726
$\text{OCH}_2\text{OCHO} = \text{HOCHO} + \text{HCO}$	3.32E+11	0.214	13,818
$\text{CH}_2\text{OCHO} = \text{HCO} + \text{CH}_2\text{O}$	8.69E+10	0.714	31,689
$\text{CH}_3\text{OCH}_2\text{O} = \text{MF} + \text{H}$	1.38E+09	1.428	12,214
$\text{C}_3\text{H}_8 + \text{OH} = n\text{-C}_3\text{H}_7 + \text{H}_2\text{O}^b$	1.05E+10	1.080	2187
$\text{C}_3\text{H}_8 + \text{OH} = i\text{-C}_3\text{H}_7 + \text{H}_2\text{O}^b$	4.67E+07	1.620	-86.8
$2\text{CH}_3\text{OCH}_2\text{O}_2 = 2\text{CH}_3\text{OCH}_2\text{O} + \text{O}_2^b$	1.60E+23	-4.500	0
$2\text{CH}_3\text{OCH}_2\text{O}_2 = \text{CH}_3\text{OCH}_2\text{OH} + \text{MF} + \text{O}_2^b$	6.84E+22	-4.500	0

^a Rates are of the form $k(T) = AT^n \exp(-E_a/RT)$ where $R = 1.9872$ cal/mol·K. Pre-exponential factor is in units of s^{-1} or $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$ depending on reaction order. MF = methyl formate.

^b Re-evaluated in this work; see text for discussion.

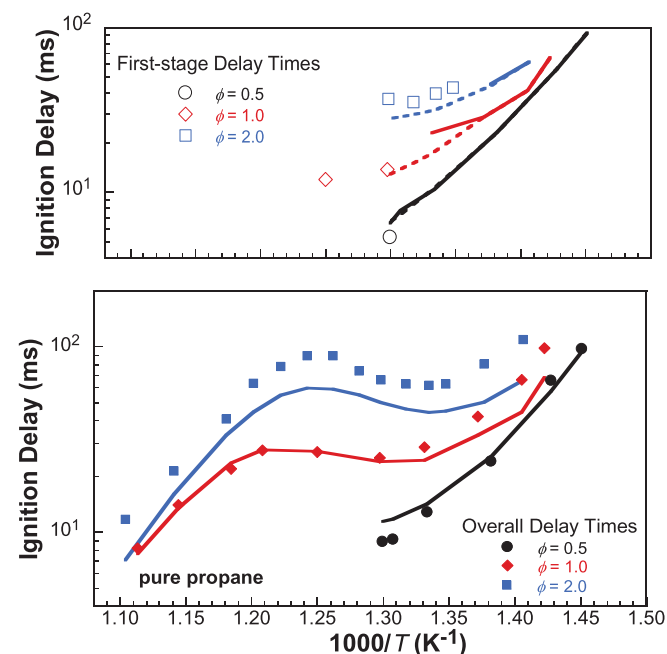


Fig. 11. Experimental propane RCM data (closed symbols: τ , open symbols: τ_1) compared with the model constructed in this work for varying ϕ at $P_c \sim 30$ bar. See Table 2 for mixture compositions. Solid lines: simulated τ or τ_1 based on dP/dt . Dashed lines: simulated τ_1 based on HRR. Predictions only span the temperature range of the experimental data because experimental pressure traces are used in simulations; see text for discussion.

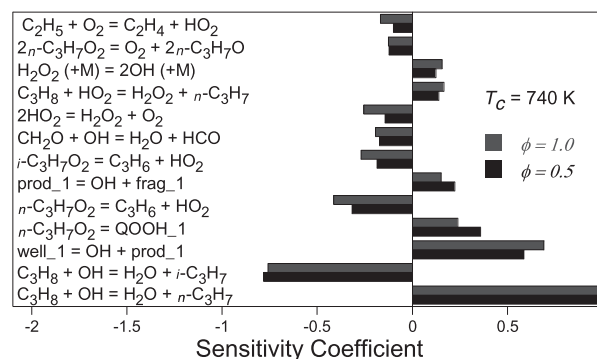
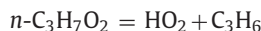


Fig. 12. Normalized sensitivity spectra for [OH] to reaction rate pre-exponential factors at the ignition delay time for the model used in this work; modeled in a batch reactor under two conditions similar to those found in the present RCM experiments ($P_c = 30$ bar, $T_c = 740$ K, pure propane mixtures of $\phi = 0.5$ and $\phi = 1.0$). Note: prod_1, frag_1, and well_1 are 2-formyl-ethyl-hydroperoxide, 2-formyl-ethoxy, and 3-hydroperoxyl-*n*-propylperoxy, respectively, as defined in [27].

pre-exponential factors for certain conditions is shown in Fig. 12. Among the reactions are the H-abstractions from propane by the OH radical, which is especially important under fuel-lean conditions. For this reason, these reactions were critically evaluated and updated in this paper, as described above. The following reactions also define the low-temperature autoignition for propane:



As seen in the sensitivity spectra of Fig. 12, the first reaction in the above list inhibits autoignition because it competes with the second O_2 addition and its subsequent chain branching pathways that produce additional OH radicals. The latter reaction promotes ignition through the net formation of two OH radicals (the decomposition of $\text{OOCH}_2\text{CH}_2\text{CH}_2\text{OOH}$ and additional decomposition of KHP to $\text{OH} + \text{OCH}_2\text{CH}_2\text{CHO}$). As explained above, the rate parameters of these reactions were adopted from the electronic structure and resulting rate theory calculations by Goldsmith et al. [27]. Mentioned in their work are the uncertainties in computed rates solely due to the 1-D treatments of loose internal degrees of freedom, which are estimated to be less than a factor of two. Given the relatively large T1 diagnostic value for many of the species in that study, the global uncertainty of the computed rate is of course larger, and at the conditions where the reactions are not in the high-pressure limit, larger uncertainties are expected. The inherently large uncertainties associated with computed rate coefficients using accurate model chemistries and state of the art rate theories can result in large global kinetic uncertainties. For this reason, reducing the uncertainties of rates common to the low-temperature oxidation of most fuel compounds requires more advanced global optimization and kinetic parametric uncertainty minimization techniques [11,122], which is beyond the scope of this paper. Thus, rates of the two reactions above were tuned within their estimated uncertain bounds, as described below, to achieve better agreement with the RCM data obtained in this paper. Figure 11 illustrates the agreement between the current RCM data and the model of this paper (see the Supplementary material, Fig. S29, for model comparisons before and after the rate adjustments) for an illustration of their overall effect on simulation results. The high sensitivity of propane ignition delay to the aforementioned rates is illustrated through the large differences in predictions between the original and updated final model. The activation energy for the rate of $n\text{-C}_3\text{H}_7\text{O}_2$ decomposition to $\text{HO}_2 + \text{C}_3\text{H}_6$ was increased by 1.0 kcal/mol, while that for the

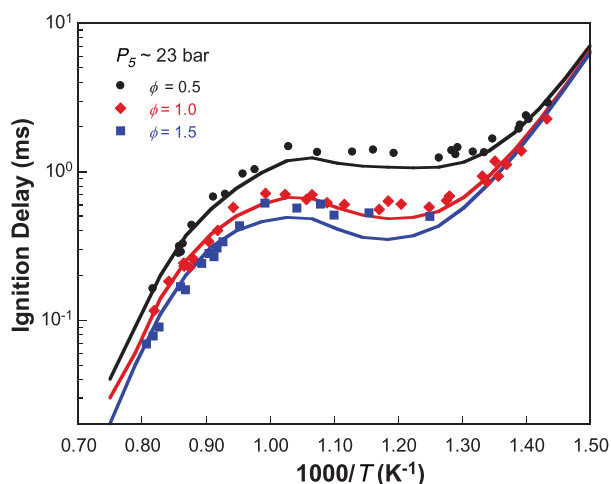


Fig. 13. Variable equivalence ratio pure DME in air ignition delay time measurements of Li et al. [23] compared with simulations using the model of this work. Black: $\phi=0.5$; red: $\phi=1.0$; blue: $\phi=1.5$; symbols: experimental data; lines: ignition delay simulation results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

decomposition of $\text{OOCH}_2\text{CH}_2\text{CH}_2\text{OOH}$ (O_2QOOH) to $\text{OH} + \text{KHP}$ was decreased by 0.5 kcal/mol. These adjustments are within the activation energy uncertainties for these two reactions. Although the resulting overall agreement is satisfactory, these rate adjustments cannot in any way reduce the uncertainty in the respective rates due to the fact that the propane ignition delay time is sensitive to a large number of reactions which also happen to have large rate parameter uncertainties. The sensitivity spectra in Fig. 12 capture this well, with many of the reactions having large sensitivity coefficients. Nonetheless, the final model performs well against the low-temperature RCM data of this paper and has been validated against a wide range of propane and DME/propane experimental targets, as shown below and in the Supplementary material. Figure 11 also presents the first-stage ignition delays (τ_1) observed experimentally for propane in this paper, with model predictions where applicable (not all simulations resulted in clearly defined first-stage ignition). A recent detailed numerical and analytical analysis of reactions driving first-stage propane ignition has been performed by Merchant et al. [28] and illustrates the importance of both OH and HO_2 to fuel consumption. In their study and at initial conditions of 10 atm and 650 K, they identify two sub-stages of exponential OH generation that occur before the overall ignition event. A first sub-stage exists where the concentration of HO_2 is high enough such that $\text{HO}_2\text{-HO}_2$ termination becomes competitive with other HO_2 reactions. This event is followed by a second sub-stage where almost all of the fuel is consumed through reaction with OH , which terminated due to depletion of the ketohydroperoxide.

4.4. DME oxidation

In order to understand DME oxidation over a wide temperature range (700–1250 K), included here are ignition delay time comparisons with the data of Li et al. [23]. As seen in Fig. 13, the agreement is excellent. The sensitivity spectra of Fig. 14 show reactions that drive ignition at 23 bar and at 1000 K for the mixture compositions in Fig. 13. Among the reactions that are shown, all of those involving DME and its large low-temperature intermediates are retained from the Burke et al. [9] model and adopted here. It is of interest to note that the KHP does not show up in this list, perhaps because it is not formed at 1000 K. Heat release rates were also computed for the conditions in Fig. 13 for $\phi=1$ in order to

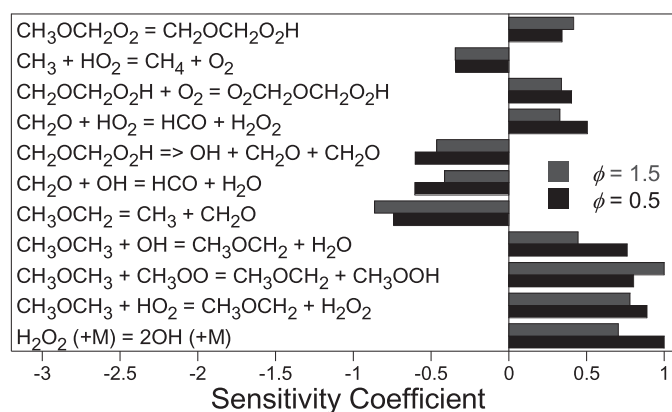
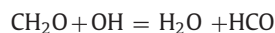
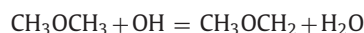
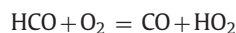
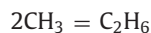
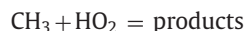
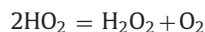


Fig. 14. Normalized sensitivity spectra for $[\text{OH}]$ to reaction rate pre-exponential factors at the ignition delay time of DME for the model used in this work and the conditions of Li et al. [23] at $T_5 = 1000 \text{ K}$ and $P_5 = 23 \text{ bar}$ for $\phi = 0.5$ (black) and $\phi = 1.5$ (grey).

better understand the reactions that drive autoignition. Between 700 and 1100 K, the following reactions are primarily responsible for heat release and listed in the order of their contribution:



Other, smaller, contributions arise from:



Thus, although many of the reactions of the DME submechanism which form and consume OH are important (and drive NTC behavior), so are the reactions of the $\text{CH}_2\text{O}/\text{HCO}$ system. At temperatures above 1100 K, CH_3OCH_2 primarily undergoes unimolecular decomposition to $\text{CH}_3 + \text{CH}_2\text{O}$, which is followed by the small molecule core chemistry of ' $\text{C}_1/\text{CH}_2\text{O}/\text{HCO}$ '.

During the low-temperature oxidation of DME, there are two dominant fates of the CH_3OCH_2 ('R') radical: unimolecular decomposition to CH_3 and CH_2O , and O_2 addition to form a stable $\text{CH}_3\text{OCH}_2\text{O}_2$ (' RO_2 ') adduct. The fraction of R that reacts to form RO_2 and eventually produce additional OH is:

$$f_{\text{R}+\text{O}_2\rightarrow\text{RO}_2} \equiv \frac{\text{rate of } \text{R} + \text{O}_2 \rightarrow \text{RO}_2}{\text{total rate of consumption of R}}$$

where the total rate of consumption of R is approximated as the sum of the rates of $\text{R} + \text{O}_2 \rightarrow \text{QOOH}$ and $\text{R} \rightarrow \text{CH}_3 + \text{CH}_2\text{O}$. In addition, a quantity defined as $f_{\text{QOOH}+\text{O}_2\rightarrow\text{O}_2\text{QOOH}}$ can be used to represent the fraction of QOOH that undergoes O_2 addition to form O_2QOOH with respect to the other two major pathways (β -scission to form $2\text{CH}_2\text{O}$ and an OH radical, or the chemically activated reaction with O_2 to form hydroperoxymethyl formate ($\text{HO}_2\text{CH}_2\text{CH}_2\text{CHO}$) + OH):

$$f_{\text{QOOH}+\text{O}_2\rightarrow\text{O}_2\text{QOOH}} \equiv \frac{\text{rate of } \text{QOOH} + \text{O}_2 \rightarrow \text{O}_2\text{QOOH}}{\text{total rate of consumption of QOOH}}$$

where the total rate of consumption of QOOH is approximated as the sum of the rates of $\text{QOOH} + \text{O}_2 \rightarrow \text{O}_2\text{QOOH}$, $\text{QOOH} + \text{O}_2 \rightarrow \text{KHP} + \text{OH}$ (chemically activated well-skipping channel), and $\text{QOOH} + \text{O}_2 \rightarrow \text{OH} + 2\text{CH}_2\text{O}$. Note that the formation of

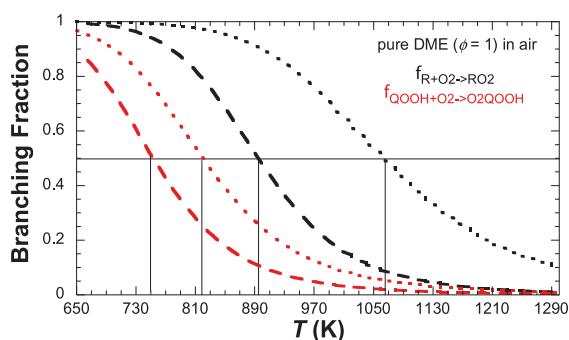


Fig. 15. Branching fractions for first and second O_2 additions to stable corresponding adducts with respect to other reactions within the pure DME/air mixture with $\phi = 1$. The dashed and dotted lines represent simulations at 1 bar and 23 bar, respectively. Red: $f_{\text{QOOH}+\text{O}_2 \rightarrow \text{O}_2\text{QOOH}}$; black: $f_{\text{R}+\text{O}_2 \rightarrow \text{RO}_2}$; horizontal line drawn at a branching fraction of 0.5; vertical lines drawn to indicate temperatures at which branching fractions equal 0.5. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a stable O_2QOOH adduct enables 2 OH radicals to be produced through its own decomposition and subsequent KHP decomposition (e.g., see Fig. 16). The branching fractions defined above are illustrated in Fig. 15 for DME in air at 1 and 23 bar (23 bar corresponding to one of the mixture compositions of Fig. 13). For the 23 bar, $\phi = 1$ case and above ~ 1050 K, the decomposition of R to CH_3 and CH_2O begins to dominate, representing the exit from the NTC region of DME. However, below ~ 1050 K (entrance to the NTC regime), $f_{\text{R}+\text{O}_2 \rightarrow \text{RO}_2} \geq 0.5$, signifying the availability of RO_2 for subsequent O_2 addition to promote chain branching. For the same pressure and mixture composition, $f_{\text{QOOH}+\text{O}_2 \rightarrow \text{O}_2\text{QOOH}} = 0.5$ around 810 K, near the shortest ignition delay times in the NTC region. As the temperature continues to decrease, the ignition delay once again becomes Arrhenius-like and driven by the kinetics of KHP decomposition.

RCM simulations using the model of this paper were also conducted for the overall and first-stage DME ignition delay time measurements of Mittal et al. [15] (see Fig. 17 for overall ignition delays, and the Supplementary material for first-stage comparisons). Good agreement is observed for overall ignition delay times at temperatures above 660 K. However, they are generally overpredicted at lower temperatures by up to a factor of two. Similarly, the first-stage ignition delay times are also overpredicted, with the largest model-experimental data discrepancy arising from fuel-rich ($\phi = 1.5$) conditions. There is also a clear predicted dependence of first- and second-stage ignition data on equivalence ratio under the lowest temperature conditions, while no corresponding dependence is observed experimentally. This is in contrast to the variable ϕ pure propane RCM measurements shown in Fig. 11, which shows a clear dependence of overall ignition delay on ϕ . Although the cause for the discrepancy between the model predictions and data of Mittal et al. [15] is unclear, some dependence of ignition delay time on ϕ is expected since a relative reduction of available O_2 decreases the flux of CH_3OCH_2 through the low-temperature chain branching pathways. In addition, we note that in the work of Mittal et al. [15], experiments were conducted with fixed [DME], while $[\text{O}_2]$ was varied to achieve desired equivalence ratios. There is a similar factor of two overprediction of delay time on the pure DME RCM results measured in this paper and shown in Fig. 20. Thus, sensitivity analyses were performed in an effort to understand what drives the predicted O_2 -dependence of ignition delay at very low temperatures. Sensitivity analyses (not shown) at 625 K and 15 bar for the mixture compositions in the right panel of Fig. 17 show that the reaction $\text{CH}_2\text{OCH}_2\text{O}_2\text{H} + \text{O}_2 = \text{O}_2\text{CH}_2\text{OCH}_2\text{O}_2\text{H}$ plays an increasingly dominant role in OH production at higher ϕ . For $\phi = 0.43$, autoignition is most sensitive to the KHP decomposition reaction, and the second O_2 addition has a sensitivity coefficient that is roughly one-half that for KHP decomposition. However, for $\phi = 1.5$, the sensitivity coefficient for the second O_2

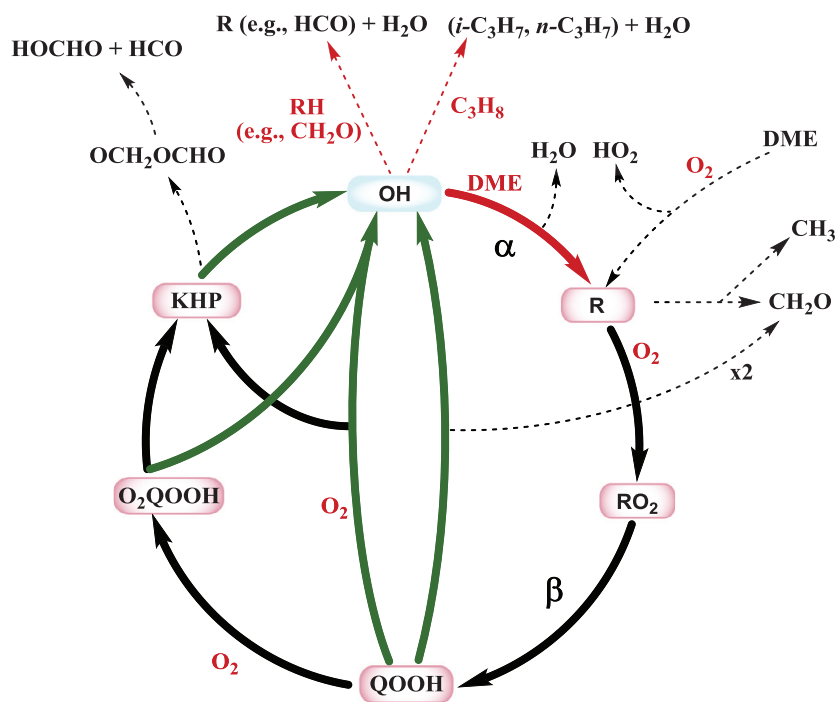


Fig. 16. Schematic of major reaction pathways during the low-temperature oxidation of DME/propane blends, with focus on DME low-temperature oxidation pathways. Solid green arrows: reactions that produce OH; solid red arrow: DME+OH H-abstraction; solid black arrows: reactions that promote the chain-branching behavior; dashed red arrows: reactions that divert the reaction pathway away from the OH gain cycle of DME; dashed black arrows: included to illustrate other product species during DME oxidation. Species shown in red text represent a reactant for that corresponding reaction. See Section 4.6 for definitions of α and β . See Fig. 3 of [28] for a similar OH cycle of propane. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

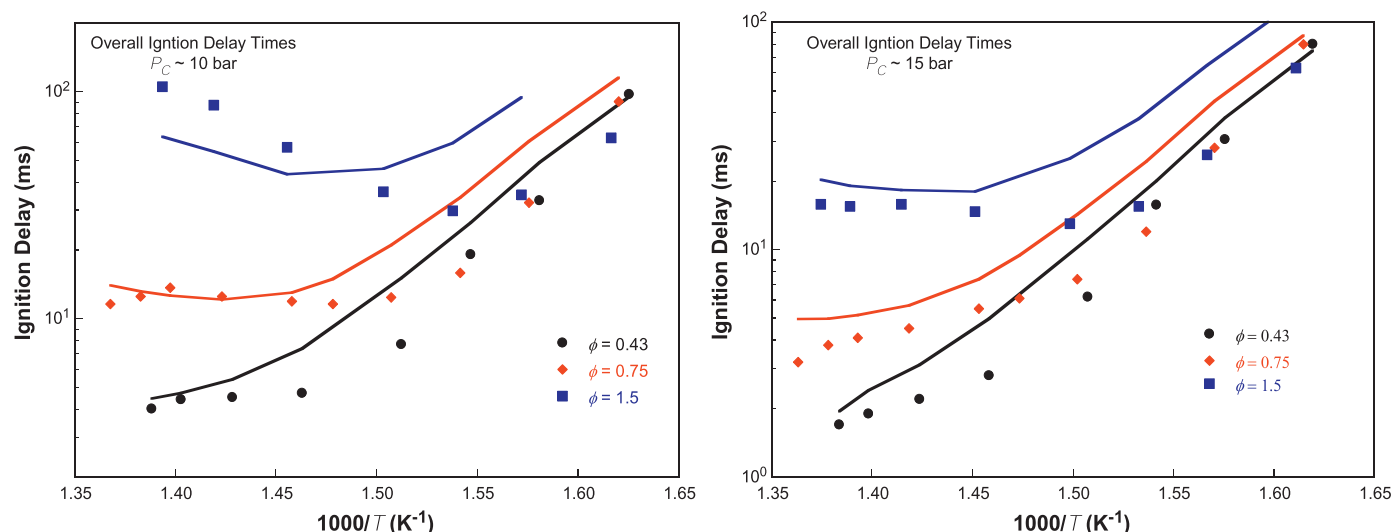


Fig. 17. Measured [15] and simulated (using the RCM model of this work) overall RCM ignition delay times at 10 bar (left) and 15 bar (right) for DME/O₂/N₂ mixtures.

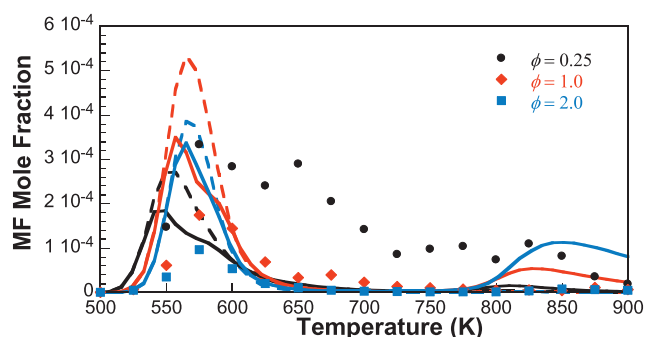


Fig. 18. Comparison of measured methyl formate (MF) species profile in a jet-stirred reactor [19] at $x_{\text{DME}} = 0.02$, $P = 10.67$ kPa, residence time = 2 s, $\phi = 0.25$, 1, and 2, with the model in this paper. Symbols: experimental data; solid lines: simulations using Model I, with 10 methyl formate reactions from Model II; dashed lines: simulations using only Model I.

addition is equal to that for the KHP decomposition, illustrating the importance of this reaction in driving DME autoignition. Thus, future work should include an improved rate coefficient for the reaction $\text{CH}_2\text{OCH}_2\text{O}_2\text{H} + \text{O}_2 = \text{O}_2\text{CH}_2\text{OCH}_2\text{O}_2\text{H}$, as well as the thermochemistry of participating species.

Rodriguez et al. [19] recently performed helium-diluted JSR experiments for three different DME equivalence ratios ($\phi = 0.25$, 1, and 2) at atmospheric pressure and in the temperature range of 500–1100 K. Along with O₂ and DME species profiles, gas chromatography was used to detect numerous additional species. Comparisons of the experimental data with simulations using the model of this paper are shown in the Supplementary material, while methyl formate species profiles comparisons are shown Fig. 18. In general, the agreement between model and data is similar to that observed in the work of Rodriguez et al. [19], but peak methyl formate (MF) mole fractions are overestimated, and acetaldehyde formation is underpredicted. Rodriguez et al. [19] noted the models of Burke et al. [9] and Zhao et al. [20] also overpredict the MF mole fractions and attributed it to the “higher reactivity” predicted by both models. Sensitivity analyses were conducted for simulated peak MF concentrations in this paper. Predicted methyl formate peak concentrations are most sensitive to three factors:

1. the thermodynamic parameters, particularly the formation enthalpy, of $\text{CH}_3\text{OCH}_2\text{O}_2$ (RO_2)

2. the self-reaction rate of $\text{CH}_3\text{OCH}_2\text{O}_2$
3. the decomposition rate of hydroperoxymethylformate/ $\text{HO}_2\text{CH}_2\text{OCHO}$ to $\text{OH} + \text{OCH}_2\text{OCHO}$.

A decrease in the standard formation enthalpy of $\text{CH}_3\text{OCH}_2\text{O}_2$ by 3 kcal/mol decreases the predicted peak MF mole fractions (at 600 K) for $\phi = 0.25$ by a factor of three, simply due to the increased endothermicity for the overall reaction sequence:



In particular, if the stability of $\text{CH}_3\text{OCH}_2\text{O}_2$ increases, the rate of the first step in the above sequence decreases (as does the rate of the minor branching channel to $\text{CH}_3\text{OCH}_2\text{OH} + \text{MF} + \text{O}_2$). This highlights the need for accurate thermochemical parameters for DME and its radical derivatives in predicting MF formation in JSR experiments. The rate of elementary steps in the above reaction sequence (discussed in detail in a previous section) has been re-evaluated and computed in this paper, and although the uncertainty in the reaction rate of $2\text{CH}_3\text{OCH}_2\text{O}_2 = 2\text{CH}_3\text{OCH}_2\text{O} + \text{O}_2$ is not known, it is likely to be quite high based on the large spread shown in the rate comparisons of Fig. 4. We also note that the predictions of this paper are in contrast to the comparisons made in the work of Rodriguez et al. [19], who predict substantially lower MF peak mole fractions. Although simulations using the model of Rodriguez et al. [19] are in good agreement with MF data for $\phi = 1$ and $\phi = 2$, the peak mole fraction is significantly underestimated for the fuel-lean condition ($\phi = 0.25$). The rate of $\text{HO}_2\text{CH}_2\text{OCHO}$ decomposition used in their model is that reported by Baulch et al. [98], which is a factor of five less than that used here at 600 K. In this paper, it is not possible to reconcile our model with the current DME RCM data without increasing the rate of this decomposition reaction from the value estimated by Baulch et al. [98]. However, this also results in an overprediction of peak MF mole fractions in the JSR experiments. High-level electronic structure and rate theory calculations are needed to give more than just an estimate for this important reaction, something outside the scope of this paper given the high multi-reference character expected in the transition state.

Peak methyl formate concentrations are also sensitive to several reactions discovered by RMG which were not included in Model I (this represents the only noticeable difference between all simulations performed using both Model I and II). Figure 18 illustrates the effect of adding these reactions on the predicted methyl

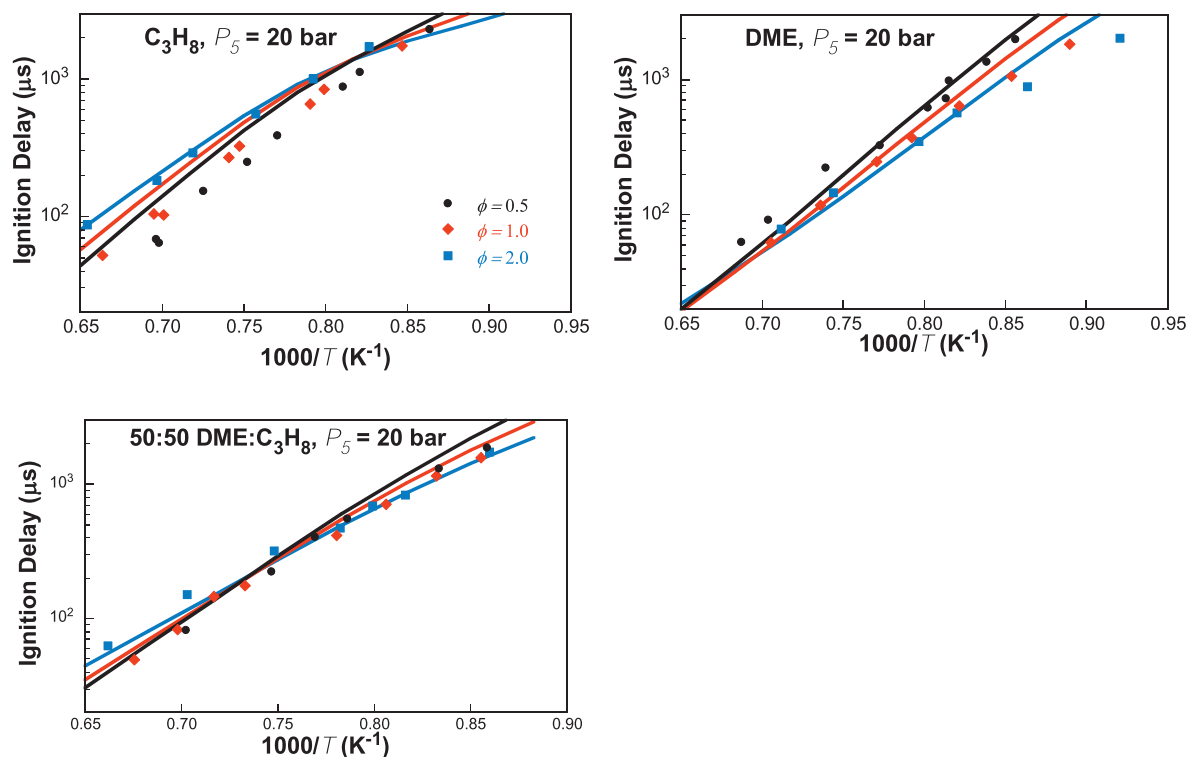


Fig. 19. Comparison of engine-relevant experimental [29] DME/propane/oxygen/argon autoignition delay times with Model I for varying mixture fractions and equivalence ratios. Lines: Model I predictions; symbols: experimental data of Hu et al. [29]. Ignition delay defined as time to maximum of $d[\text{OH}]/dt$.

formate peaks. The greatest difference occurs for $\phi = 1$; inclusion of the RMG generated reactions reduces the peak methyl formate by $\sim 40\%$. In addition, the RMG generated reactions promote the production of methyl formate above 800 K, although only for $\phi = 1$ and $\phi = 2$. Because the RMG-generated reactions have a substantial influence on these predictions, they are included in the final model. These 10 reactions are primarily metathesis reactions between DME and its large radical derivatives.

4.5. DME/propane binary fuel mixtures

Although the focus of this paper primarily lies in the lower temperature ($T < 1000$ K) regime, selected shock tube simulations were conducted at high temperatures for DME, propane, and their mixtures to verify that the final model performs well under such conditions. Figure 19 illustrates comparisons with the recent work of Hu et al. [29], which is the only DME/propane blended shock tube study to date. In addition, many other experimental targets included in Table 1 were simulated with Model I in this paper; comparisons can be found in the Supplementary material.

Figure 20 illustrates the influence of DME on the autoignition of propane in their binary mixtures in RCM experiments conducted in this paper. As described above, a total of three reactions were adjusted in the model of this paper to achieve better agreement with the RCM data (see the Supplementary material for comparisons of the model before and after rate adjustments). The resulting model performs reasonably well against the overall ignition delay times, capturing the fuel blend effects. Even a 10% addition by mole fraction of DME to propane reduces the ignition delay times by a factor of three in the NTC region and a factor of two at both the highest and lowest temperatures studied. Also illustrated in Fig. 20 are first-stage ignition delay times. The overall model-experimental data agreement is satisfactory. Sensitivity analyses were carried out for both $[\text{DME}]$ and $[\text{OH}]$ at the first-

stage for the 100% DME mixture at 770 K and 30 bar. As expected, the reactions of the low-temperature O_2 addition sequences and subsequent chain branching pathways, in addition to $\text{DME} + \text{O}_2 = \text{CH}_3\text{OCH}_2 + \text{HO}_2$, dominate OH production and DME consumption. $\text{CH}_3\text{OCH}_3 + \text{HO}_2 = \text{CH}_3\text{OCH}_2 + \text{H}_2\text{O}_2$ is also shown to play a role, albeit a minor one.

Figure S31 compares the DME/propane blended RCM data of this paper to simulation results provided by other relevant models available in the literature. The model of Rodriguez et al. [19] significantly overestimates the ignition delay time of the pure propane and 10:90 DME:propane mixtures. DME ignition delay times predicted using the model of Liu et al. [12] and Wang et al. [42] are in good agreement with the current RCM data, primarily because the KHP decomposition rates is so high (see Fig. 2). Recall that this is also the reason the model overpredicts MF peak mole fractions measured in the JSR experiments of Rodriguez et al. [19]. Thus, there is an apparent trade-off between reconciling model predictions with low-temperature shock tube and RCM data and accurately capturing the MF JSR species profiles.

As seen in the sensitivity spectrum for a binary DME/propane mixture (Fig. 21), the decomposition of hydroperoxymethylformate to produce OH drives overall consumption of both DME and propane below 700 K. In addition, the sensitivity spectrum of propane is qualitatively the same as that for DME, illustrating the strong coupling of the two fuels as well as the promoting effect of DME on propane consumption. However, sensitivities of fuel concentration to H-abstraction by OH from both parent fuels are of opposing magnitude. This is due to the different overall reactivity of propane and DME under these conditions, where the low-temperature chain branching cycle of DME is active, but that for propane is not. Thus, the reaction of $\text{C}_3\text{H}_8 + \text{OH}$ to *i*-propyl is in effect a radical sink under the conditions shown in Fig. 21. As discussed by Merchant et al. [28], this is due to the inability of *i*-propyl to undergo OH producing chain branching reactions. Instead, it goes on to produce $\text{HO}_2 + \text{propene}$.

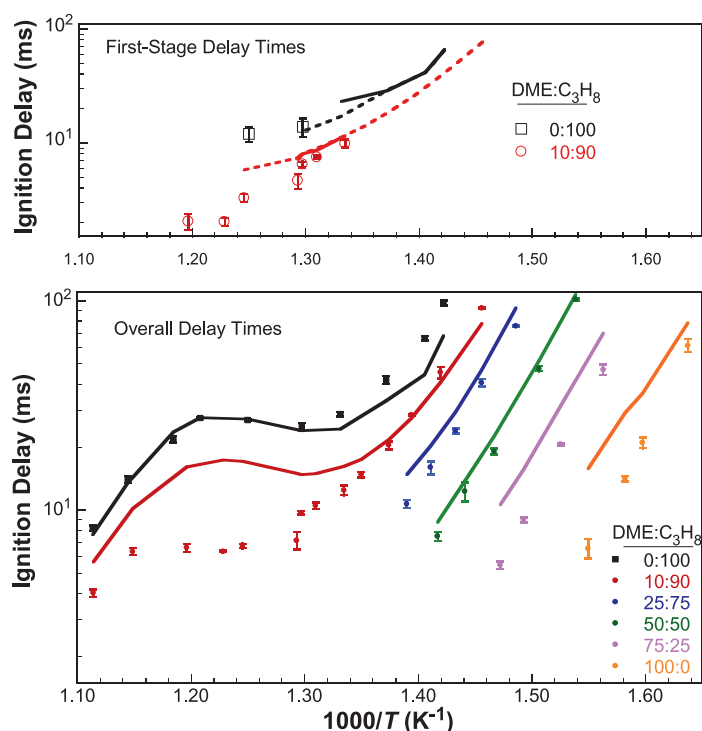


Fig. 20. Comparison of RCM data with present model for varying DME/propane blend ratios by mole ($P_c \sim 30$ bar, $\phi = 1.0$). The blending ratio is defined within plot; solid lines: simulated overall and first-stage ignition delays with final model of this work based on dp/dt ; open circles: first-stage ignition delay time (τ_1) measurements of this work; dashed lines: simulation of τ_1 based on the HRR. Note that not all simulations resulted in clearly defined first-stage ignition.

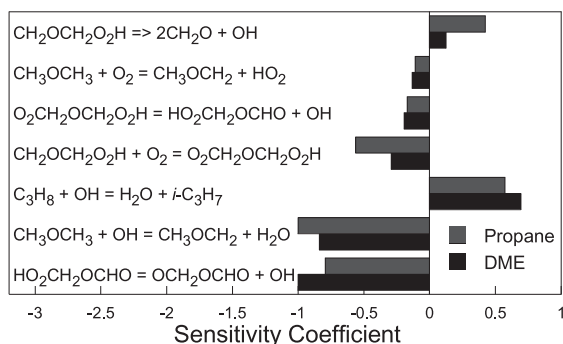


Fig. 21. Normalized sensitivity spectra for [DME] and [Propane] at 50% propane consumption to reaction rate pre-exponential factors for the model used in this work, modeled in a batch reactor under conditions similar to those found in the present RCM experiments ($T=690$ K, $P=30$ bar, 25:75 DME:propane mixture in air at $\phi=1$).

4.6. Propane combustion enhancement via DME

Under the fuel blend conditions studied here, propane combustion is promoted as a result of low-temperature DME oxidation through the schematic of Fig. 16. DME reacts with molecular oxygen to produce the CH_3OCH_2 and HO_2 radicals, the former of which then engages the low-temperature autoignition sequence of DME through O_2 addition to $\text{CH}_3\text{OCH}_2\text{O}_2$. A net gain of OH radicals results in H-abstraction from propane, generating *i*-propyl and *n*-propyl radicals, the latter of which engages the low-temperature autoignition chemistry of propane. For all binary mixtures studied in this paper, we observed similar rates of decay of both DME and propane, within 3% up to the ignition delay time. This is due to a balance in the rates of H-abstraction by HO_2 and OH from the primary fuel molecules. Over the whole temperature range, the total rate of H-abstraction by OH is a factor two higher for DME than

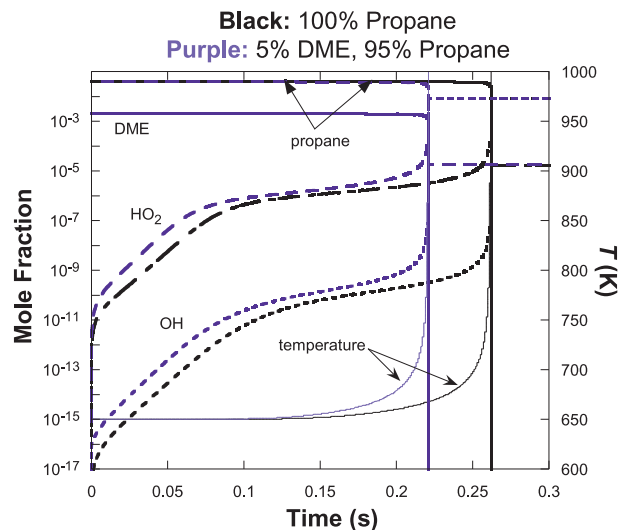


Fig. 22. Temperature, OH, HO_2 , and fuel species profiles for a CONV reactor simulation of a stoichiometric 100% propane/air mixture (black lines) and a 5:95 DME:propane blend in air (purple lines) at an initial temperature and pressure of 650 K and 30 atm, respectively. Simulations are conducted using the model of this work. Temperature ordinate maximum is set to 1000 K for easier viewing of corresponding temperature profiles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for propane. However, for the intermediate to high temperatures encountered in this study, the rate of H-abstraction by HO_2 is similar between the fuels, with abstraction from propane increasing in importance as the temperature increases. Since there is a much greater concentration of HO_2 than OH (by orders of magnitude; e.g., see Fig. 22), the rates of consumption of DME and propane are approximately the same throughout the ignition delay.

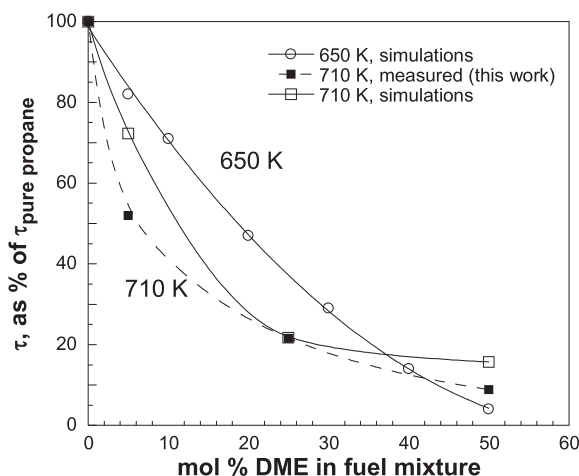


Fig. 23. Influence of DME addition on binary DME/propane/air mixtures. DME substitution by mol % to an undiluted stoichiometric propane/air mixture ($\phi = 1$, $P = 30$ atm) for two temperatures. Filled squares: τ values from the RCM measurements of this work at 710 K with corresponding RCM-simulated values (open squares); open circles: CONV simulations of τ at 650 K (fits to each data set provided to guide the eye).

Illustrated in Fig. 22 is the promoting effect of DME on propane autoignition: a 5:95 DME:propane blend ratio (by mole) contains enough DME to reduce propane autoignition by 20% under these conditions. The binary blend composition was created by replacing 5% (by mole) of the propane in a stoichiometric propane/air mixture with DME. We also tested the relative influence of $O_2 + DME = HO_2 + CH_3OCH_2$ on the promoting effect of DME on propane. This reaction was removed from the model, and simulations similar to those in Fig. 22 were performed, with only a slight increase in the overall ignition delay time for the 5:95 DME:propane mixture. Thus, the promoting effect of DME on propane autoignition is dominated by OH and HO_2 .

Figure 23 illustrates the nonlinear promoting effect of DME on propane for stoichiometric binary DME/propane mixtures under various conditions. Due to the existence of an NTC-regime for this binary fuel blend, the promoting effect under varying initial conditions is neither exponential nor second order in DME concentration. Clearly though, nonlinear trends are predicted and experimentally observed for the 710 K case. This temperature in particular was chosen because it is the only temperature where overall ignition delay times could be measured in this paper for all fuel blends up to 50:50 DME:propane. The strong promoting effect of DME on overall autoignition even in low quantities (also illustrated through the 10:90 mixture of Figure 16) illustrates why this compound has been proposed as a fuel additive in automotive engines.

Overall OH gain (for $T < 700$ K) is another way to understand the promoting effect of DME on propane. OH gain can be defined as the fraction of OH radicals created compared to those consumed in the low-temperature OH cycle of a given compound. We adopt similar conventions for OH gain of DME as for propane in the work of Merchant et al. [28], who studied OH chain propagation before overall ignition events, where O_2 and fuel concentrations are effectively constant; this, in combination with a negligible temperature rise, enables a simple analytical solution to the time-evolution of [OH]. Two pressure- and temperature-dependent branching ratios are defined in Fig. 16:

$$\alpha = \frac{\text{net rate of } OH + RH \rightarrow R + H_2O}{\text{total consumption rate of OH}}$$

$$\beta = \frac{\text{net rate of } RO_2 \rightarrow QOOH}{\text{total consumption rate of } RO_2}$$

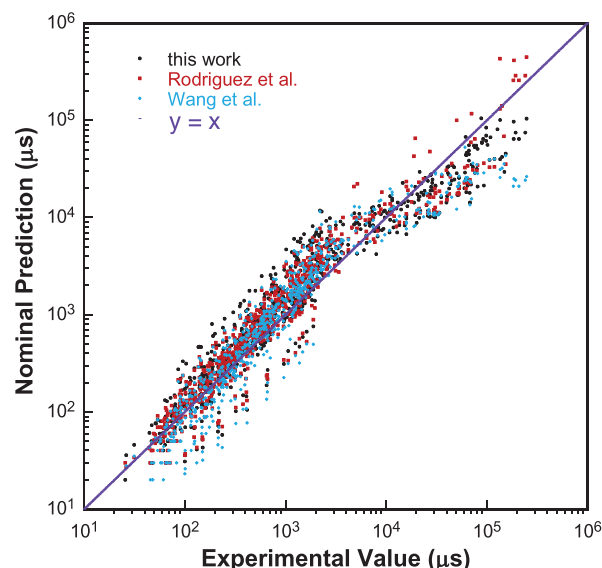


Fig. 24. Parity plot of all experimental shock tube data in this paper and the Supplementary material ([9,21,22,23,29,43,44,47]) and the RCM data from this work with the corresponding predicted ignition delay time values from the model in this work (black circles), the model from Rodriguez et al. [19] (red squares), and the model from Wang et al. [42] (blue diamonds). Purple line: $y = x$, for visual comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Using the values above, the total OH gain for DME can be derived:

$$f_{OH,DME} = \alpha\beta\theta$$

where θ represents the OH gain from the fate of the QOOH species. We note the above expression implicitly assumes that the branching ratio of $R + O_2$ to the total consumption rate of R is one. For the conditions described above, $\alpha\beta = 1$ because every OH generated goes on to attack DME, and the RO_2 -QOOH isomerization dominates. If the QOOH decomposition to $OH + 2CH_2O$ pathway is negligible, $\theta \sim 2$ since both $QOOH + O_2$ channels can result in the production of 2 OH radicals. The total OH gain of DME then simplifies to $f_{OH,DME} \sim 2$ at 650 K. A similar analysis has been conducted for propane, with an OH gain in the region of 500–700 K of $f_{OH,C_3H_8} = 3\alpha\beta$ [28]. As discussed by Merchant et al. [28], the propane KHP chemical amplifier has an OH gain of about 1.5 (i.e., each OH radical produced results in another 1.5 OH radicals produced). As temperature increases, the OH gain drops primarily due to the increased rate of $CH_3CH_2CH_2O_2 = HO_2 + \text{propene}$. Above about 700 K, the propane KHP cycle no longer amplifies [OH]. Similarly, $f_{OH,DME}$ drops with increasing temperature, primarily due to the increased rate of the reaction $CH_2OCH_2OOH = 2CH_2O + OH$ which competes with θ . For DME/propane blends, the overall gain is an average of the pure molecule gains, weighted by the rates of H-abstraction by OH on each parent fuel compound:

$$f_{OH,total} = \frac{k_{OH+DME}x_{DME}f_{OH,DME} + k_{OH+C_3H_8}x_{C_3H_8}f_{OH,C_3H_8}}{k_{OH+DME}x_{DME} + k_{OH+C_3H_8}x_{C_3H_8}}$$

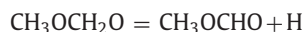
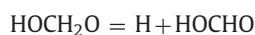
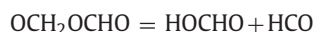
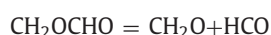
In the above equation, $k_{OH+C_3H_8}$ represents the rate for the reaction that produces n -propyl + H_2O . Consider a system where $x_{DME} = x_{C_3H_8} = 0.5$ and $k_{OH+DME} = 2k_{OH+C_3H_8}$ (over the temperature range of this paper, the total rate of H-abstraction by OH is a factor two higher for DME than for propane). For these conditions, the overall gain of the binary fuel mixture becomes $f_{OH,total} = 2/3f_{OH,DME} + 1/3f_{OH,C_3H_8}$. At a temperature of 650 K, $f_{OH,DME} \sim 2$ as described above, $f_{OH,C_3H_8} \leq 1.5$, and thus $f_{OH,total} \leq 1.5$. The above equation simplifies to $f_{OH,DME}$ or f_{OH,C_3H_8} for unity mole fractions of x_{DME} or $x_{C_3H_8}$, respectively.

4.7. Overall model performance

A parity plot of the experimental ignition delay time data points taken from the shock tube experiments listed in Table 1 and the RCM experiments from this paper is shown in Fig. 24 for the model presented in this paper, the Rodriguez et al. model [19], and the Wang et al. model [42]. Comparisons with the Burke et al. [9] model are not shown since much of the DME chemistry in this model is directly adopted into the model of this paper. In general, all models perform equally well. It is interesting to note the general model-data discrepancy at very long overall ignition delay times, where achieving constant volume reactor conditions becomes a challenge (or, simply not possible), as recently discussed by Hanson et al. [87]. Additional experimental targets (e.g., species profiles behind reflected shock waves, perfectly stirred reactor data) were also considered and are included in the Supplementary material. Finally, an IPython [123] notebook is provided in the Supplementary material to ensure partial reproducibility of a number of the figures in this paper, with simulations performed using Cantera [84].

5. Conclusions

A model for the oxidation of DME/propane binary fuel mixtures has been developed and validated against a wide range of experimental targets. A number of recent advances in DME and propane combustion have been compiled and included in this model. In particular, the pressure-dependent unimolecular decomposition of CH_3OCH_2 [9] has been included from other work and the concerted elimination from OCH_2OCHO to $\text{HOCHO} + \text{HCO}$ [42] along with an accurate evaluation for the reaction and branching of $\text{C}_3\text{H}_8 + \text{OH}$, a re-evaluation of $2\text{CH}_3\text{OCH}_2\text{O}_2 = \text{products}$, and accurate computation of pressure-dependent unimolecular decomposition rates for several radical intermediates of DME:



In addition to being radical intermediates during low-temperature DME oxidation, some of the above species are directly relevant to the combustion of methyl formate, acetone, and methylene glycol.

New RCM data have been obtained at 30 bar for $T_c = 600\text{--}900\text{ K}$ for pure propane mixtures as well as a range of DME/propane blends. The model constructed in this paper performs well against the RCM data as well as other literature experimental data. Both the RCM data and model illustrate the significant promoting effect DME has on propane autoignition.

The Reaction Mechanism Generator (RMG) tool was additionally utilized in an effort to discover cross-reactions between large radical intermediates of both parent fuel compounds. Although RMG generated over 3000 additional reactions including many cross-reactions, they did not have an effect on most simulation results. The important cross-reactions that do occur for this binary fuel blend are through the OH and HO_2 radicals, suggesting that binary fuel models designed for dual fuel purposes may be assembled simply based on combinations of their separate submechanisms. This observation, in combination with advanced model merging tools [97] that can identify and eliminate redundant species and/or reactions, may make model generation for future binary fuel mixtures a facile task (assuming the independent submechanisms are reliable).

Acknowledgments

The work at the University of Connecticut was supported by the National Science Foundation under Grant No. CBET-1402231.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2016.02.021.

References

- [1] C. Arcoumanis, C. Bae, R. Crookes, E. Kinoshita, The potential of di-methyl ether (DME) as an alternative fuel for compression-ignition engines: A review, *Fuel* 87 (2008) 1014–1030.
- [2] T. Kitamura, T. Ito, J. Senda, H. Fujimoto, Mechanism of smokeless diesel combustion with oxygenated fuels based on the dependence of the equivalence ratio and temperature on soot particle formation, *Int. J. Engine Res.* 3 (2002) 223–248.
- [3] X. Lu, D. Han, Z. Huang, Fuel design and management for the control of advanced compression-ignition combustion modes, *Prog. Energy Combust. Sci.* 37 (2011) 741–783.
- [4] B. Prabhakar, S. Jayaraman, R. Vander Wal, A. Boehman, Experimental studies of high efficiency combustion with fumigation of dimethyl ether and propane into diesel engine intake air, *J. Eng. Gas Turbines Power* 137 (2015).
- [5] S.L. Kokjohn, R.M. Hanson, D.A. Splitter, R.D. Reitz, Fuel reactivity controlled compression ignition (RCCI): a pathway to controlled high-efficiency clean combustion, *Int. J. Engine Res.* 12 (2011) 209–226.
- [6] D. Splitter, R. Hanson, S. Kokjohn, R.D. Reitz, Reactivity controlled compression ignition (RCCI) heavy-duty engine operation at mid-and high-loads with conventional and alternative fuels, SAE Technical Paper (2011).
- [7] M.H. Morsy, Ignition control of methane fueled homogeneous charge compression ignition engines using additives, *Fuel* 86 (2007) 533–540.
- [8] Z. Chen, X. Qin, Y. Ju, Z. Zhao, M. Chaos, F.L. Dryer, High temperature ignition and combustion enhancement by dimethyl ether addition to methane-air mixtures, *Proc. Combust. Inst.* 31 (2007) 1215–1222.
- [9] U. Burke, K.P. Somers, P. O'Toole, C.M. Zinner, N. Marquet, G. Bourque, E.L. Petersen, W.K. Metcalfe, Z. Serinyel, H.J. Curran, An ignition delay and kinetic modeling study of methane, dimethyl ether, and their mixtures at high pressures, *Combust. Flame* 162 (2015) 315–330.
- [10] M. Mehl, W.J. Pitz, C.K. Westbrook, H.J. Curran, Kinetic modeling of gasoline surrogate components and mixtures under engine conditions, *Proc. Combust. Inst.* 33 (2011) 193–200.
- [11] A.S. Tomlin, E. Agbro, V. Nevrlý, J. Dlabka, M. Vašínek, Evaluation of combustion mechanisms using global uncertainty and sensitivity analyses: a case study for low-temperature dimethyl ether oxidation, *Int. J. Chem. Kinet.* 46 (2014) 662–682.
- [12] D. Liu, J. Santner, C. Togbé, D. Felsmann, J. Koppmann, A. Lackner, X. Yang, X. Shen, Y. Ju, K. Kohse-Höinghaus, Flame structure and kinetic studies of carbon dioxide-diluted dimethyl ether flames at reduced and elevated pressures, *Combust. Flame* 160 (2013) 2654–2668.
- [13] X.L. Zheng, T.F. Lu, C.K. Law, C.K. Westbrook, H.J. Curran, Experimental and computational study of nonpremixed ignition of dimethyl ether in counter-flow, *Proc. Combust. Inst.* 30 (2005) 1101–1109.
- [14] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modeling study of $\text{C}_1\text{--C}_2$ hydrocarbon and oxygenated fuels, *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [15] G. Mittal, M. Chaos, C.-J. Sung, F.L. Dryer, Dimethyl ether autoignition in a rapid compression machine: Experiments and chemical kinetic modeling, *Fuel Process. Technol.* 89 (2008) 1244–1254.
- [16] A.J. Eskola, S.A. Carr, R.J. Shannon, B. Wang, M.A. Blitz, M.J. Pilling, P.W. Seakins, Analysis of the kinetics and yields of OH radical production from the $\text{CH}_3\text{OCH}_2 + \text{O}_2$ reaction in the temperature range 195–650 K: an experimental and computational study, *J. Phys. Chem. A* 118 (2014) 6773–6788.
- [17] S. Fischer, F. Dryer, H. Curran, The reaction kinetics of dimethyl ether. I: high-temperature pyrolysis and oxidation in flow reactors, *Int. J. Chem. Kinet.* 32 (2000) 713–740.
- [18] H. Curran, S. Fischer, F. Dryer, The reaction kinetics of dimethyl ether. II: low-temperature oxidation in flow reactors, *Int. J. Chem. Kinet.* 32 (2000) 741–759.
- [19] A. Rodriguez, O. Frotter, O. Herbinet, R. Fournet, R. Bounaceur, C. Fittschen, F. Battin-Leclerc, Experimental and modeling investigation of the low-temperature oxidation of dimethyl ether, *J. Phys. Chem. A* 28 (2015) 7905–7923.
- [20] Z. Zhao, M. Chaos, A. Kazakov, F.L. Dryer, Thermal decomposition reaction and a comprehensive kinetic model of dimethyl ether, *Int. J. Chem. Kinet.* 40 (2008) 1–18.
- [21] P. Dagaut, C. Daly, J.M. Simmie, M. Cathonnet, The oxidation and ignition of dimethylether from low to high temperature (500–1600 K): experiments and kinetic modeling, *Symp. (Int.) Combust.* 27 (1998) 361–369.

- [22] U. Pfahl, K. Fieweger, G. Adomeit, Self-ignition of diesel-relevant hydrocarbon-air mixtures under engine conditions, *Symp. (Int.) Combust.* 26 (1996) 781–789.
- [23] Z. Li, W. Wang, Z. Huang, M.A. Oehlschlaeger, Dimethyl ether autoignition at engine-relevant conditions, *Energy Fuels* 27 (2013) 2811–2817.
- [24] J.M. Simmie, Detailed chemical kinetic models for the combustion of hydrocarbon fuels, *Prog. Energy Combust. Sci.* 29 (2003) 599–634.
- [25] C.K. Westbrook, F.L. Dryer, Chemical kinetic modeling of hydrocarbon combustion, *Prog. Energy Combust. Sci.* 10 (1984) 1–57.
- [26] S. Gallagher, H. Curran, W. Metcalfe, D. Healy, J. Simmie, G. Bourque, A rapid compression machine study of the oxidation of propane in the negative temperature coefficient regime, *Combust. Flame* 153 (2008) 316–333.
- [27] C.F. Goldsmith, W.H. Green, S.J. Klippenstein, Role of $O_2 + QOOH$ in low-temperature ignition of propane. 1. temperature and pressure dependent rate coefficients, *J. Phys. Chem. A* 116 (2012) 3325–3346.
- [28] S.S. Merchant, C.F. Goldsmith, A.G. Vandeputte, M.P. Burke, S.J. Klippenstein, W.H. Green, Understanding low-temperature first-stage ignition delay: propane, *Combust. Flame* 162 (2015) 3658–3673.
- [29] E. Hu, Z. Zhang, L. Pan, J. Zhang, Z. Huang, Experimental and modeling study on ignition delay times of dimethyl ether/propane/oxygen/argon mixtures at 20 bar, *Energy Fuels* 27 (2013) 4007–4013.
- [30] T. Yamada, J.W. Bozzelli, T.H. Lay, Comparisons of CBS-q and G2 calculations on thermodynamic properties, transition states, and kinetics of dimethyl-ether+O reaction system, *Int. J. Chem. Kinet.* 32 (2000) 435–452.
- [31] J.W. Allen, C.F. Goldsmith, W.H. Green, Automatic estimation of pressure-dependent rate coefficients, *Phys. Chem. Chem. Phys.* 14 (2012) 1131–1155.
- [32] P. Dagaut, J.-C. Boettner, M. Cathonnet, Chemical kinetic study of dimethylether oxidation in a jet stirred reactor from 1 to 10 atm: experiments and kinetic modeling, *Symp. (Int.) Combust.* 26 (1996) 627–632.
- [33] N.L. Le Tan, M. Djehiche, C.D. Jain, P. Dagaut, G. Dayma, Quantification of HO_2 and other products of dimethyl ether oxidation (H_2O_2 , H_2O , and CH_2O) in a jet-stirred reactor at elevated temperatures by low-pressure sampling and continuous-wave cavity ring-down spectroscopy, *Fuel* 158 (2015) 248–252.
- [34] M.U. Alzueta, J. Muro, R. Bilbao, P. Glarborg, Oxidation of dimethyl ether and its interaction with nitrogen oxides, *Isr. J. Chem.* 39 (1999) 73–86.
- [35] I. Liu, N.W. Cant, J.H. Bromly, F.J. Barnes, P.F. Nelson, B.S. Haynes, Formate species in the low-temperature oxidation of dimethyl ether, *Chemosphere* 42 (2001) 583–589.
- [36] H. Guo, W. Sun, F.M. Haas, T. Farouk, F.L. Dryer, Y. Ju, Measurements of HO in low temperature dimethyl ether oxidation, *Proc. Combust. Inst.* 34 (2013) 573–581.
- [37] F. Herrmann, B. Jochim, P. Oßwald, L. Cai, H. Pitsch, K. Kohse-Höinghaus, Experimental and numerical low-temperature oxidation study of ethanol and dimethyl ether, *Combust. Flame* 161 (2014) 384–397.
- [38] F. Herrmann, P. Oßwald, K. Kohse-Höinghaus, Mass spectrometric investigation of the low-temperature dimethyl ether oxidation in an atmospheric pressure laminar flow reactor, *Proc. Combust. Inst.* 34 (2013) 771–778.
- [39] A. Schönborn, P. Sayad, A.A. Konnov, J. Klingmann, Autoignition of dimethyl ether and air in an optical flow-reactor, *Energy Fuels* 28 (2014) 4130–4138.
- [40] T. Wada, A. Sudholt, H. Pitsch, N. Peters, Analysis of first stage ignition delay times of dimethyl ether in a laminar flow reactor, *Combust. Theory Model* 17 (2013) 906–936.
- [41] N. Kurimoto, B. Brumfield, X. Yang, T. Wada, P. Diévert, G. Wysocki, Y. Ju, Quantitative measurements of HO_2/H_2O_2 and intermediate species in low and intermediate temperature oxidation of dimethyl ether, *Proc. Combust. Inst.* 35 (2015) 457–464.
- [42] Z. Wang, X. Zhang, L. Xing, L. Zhang, F. Herrmann, K. Moshhammer, F. Qi, K. Kohse-Höinghaus, Experimental and kinetic modeling study of the low- and intermediate-temperature oxidation of dimethyl ether, *Combust. Flame* 162 (2015) 1113–1125.
- [43] R.D. Cook, D.F. Davidson, R.K. Hanson, Shock tube measurements of ignition delay times and OH time-histories in dimethyl ether oxidation, *Proc. Combust. Inst.* 32 (2009) 189–196.
- [44] C. Tang, L. Wei, J. Zhang, X. Man, Z. Huang, Shock tube measurements and kinetic investigation on the ignition delay times of methane/dimethyl ether mixtures, *Energy Fuels* 26 (2012) 6720–6728.
- [45] S.H. Pyun, W. Ren, K.-Y. Lam, D.F. Davidson, R.K. Hanson, Shock tube measurements of methane, ethylene and carbon monoxide time-histories in DME pyrolysis, *Combust. Flame* 160 (2013) 747–754.
- [46] L. Pan, E. Hu, J. Zhang, Z. Zhang, Z. Huang, Experimental and kinetic study on ignition delay times of DME/ $H_2/O_2/Ar$ mixtures, *Combust. Flame* 161 (2014) 735–747.
- [47] L. Pan, E. Hu, Z. Tian, F. Yang, Z. Huang, Experimental and kinetic study on ignition delay times of dimethyl ether at high temperatures, *Energy Fuels* 29 (2015) 3495–3506.
- [48] A. McIlroy, T.D. Hain, H.A. Michelsen, T.A. Cool, A laser and molecular beam mass spectrometer study of low-pressure dimethyl ether flames, *Proc. Combust. Inst.* 28 (2000) 1647–1653.
- [49] E.W. Kaiser, T.J. Wallington, M.D. Hurley, J. Platz, H.J. Curran, W.J. Pitz, C.K. Westbrook, Experimental and modeling study of premixed atmospheric-pressure dimethyl ether-air flames, *J. Phys. Chem. A* 104 (2000) 8194–8206.
- [50] T.A. Cool, J. Wang, N. Hansen, P.R. Westmoreland, F.L. Dryer, Z. Zhao, A. Kazakov, T. Kasper, K. Kohse-Höinghaus, Photoionization mass spectrometry and modeling studies of the chemistry of fuel-rich dimethyl ether flames, *Proc. Combust. Inst.* 31 (2007) 285–293.
- [51] J. Wang, U. Struckmeier, B. Yang, T.A. Cool, P. Osswald, K. Kohse-Höinghaus, T. Kasper, N. Hansen, P.R. Westmoreland, Isomer-specific influences on the composition of reaction intermediates in dimethyl ether/propene and ethanol/propene flame, *J. Phys. Chem. A* 112 (2008) 9255–9265.
- [52] H. Xu, C. Yao, T. Yuan, K. Zhang, H. Guo, Measurements and modeling study of intermediates in ethanol and dimethyl ether low-pressure premixed flames using synchrotron photoionization, *Combust. Flame* 158 (2011) 1673–1681.
- [53] Y.L. Wang, A.T. Holley, C. Ji, F.N. Egolopoulos, T.T. Tsotsis, H.J. Curran, Propagation and extinction of premixed dimethyl-ether/air flames, *Proc. Combust. Inst.* 32 (2009) 1035–1042.
- [54] C.A. Daly, J.M. Simmie, J. Würmel, N. Djeballi, C. Paillard, Burning velocities of dimethyl ether and air, *Combust. Flame* 125 (2001) 1329–1340.
- [55] X. Qin, Y. Ju, Measurements of burning velocities of dimethyl ether and air premixed flames at elevated pressures, *Proc. Combust. Inst.* 30 (2005) 233–240.
- [56] Z. Huang, Q. Wang, J. Yu, Y. Zhang, K. Zeng, H. Miao, D. Jiang, Measurement of laminar burning velocity of dimethyl ether–air premixed mixtures, *Fuel* 86 (2007) 2360–2366.
- [57] Z. Chen, L. Wei, Z. Huang, H. Miao, X. Wang, D. Jiang, Measurement of laminar burning velocities of dimethyl ether–air premixed mixtures with N_2 and CO_2 dilution, *Energy Fuels* 23 (2009) 735–739.
- [58] J. De Vries, W.B. Lowry, Z. Serinyel, H.J. Curran, E.L. Petersen, Laminar flame speed measurements of dimethyl ether in air at pressures up to 10 atm, *Fuel* 90 (2011) 331–338.
- [59] G. Mittal, C.-J. Sung, A rapid compression machine for chemical kinetics studies at elevated pressures and temperatures, *Combust. Sci. Technol.* 179 (2007) 497–530.
- [60] G. Mittal, C.-J. Sung, Aerodynamics inside a rapid compression machine, *Combust. Flame* 145 (2006) 160–180.
- [61] D. Lee, S. Hochgreb, Rapid compression machines: Heat transfer and suppression of corner vortex, *Combust. Flame* 114 (1998) 531–545.
- [62] B.W. Weber, C.-J. Sung, M.W. Renfro, On the uncertainty of temperature estimation in a rapid compression machine, *Combust. Flame* 162 (2015) 2518–2528.
- [63] T.B. Adler, G. Knizia, H.-J. Werner, A simple and efficient CCSD (T)-F12 approximation, *J. Chem. Phys.* 127 (2007) 221106.
- [64] T.B. Adler, H.-J. Werner, Local explicitly correlated coupled-cluster methods: efficient removal of the basis set incompleteness and domain errors, *J. Chem. Phys.* 130 (2009) 241101.
- [65] T.B. Adler, H.-J. Werner, F.R. Manby, Local explicitly correlated second-order perturbation theory for the accurate treatment of large molecules, *J. Chem. Phys.* 130 (2009) 054106.
- [66] G. Knizia, T.B. Adler, H.-J. Werner, Simplified CCSD (T)-F12 methods: theory and benchmarks, *J. Chem. Phys.* 130 (2009) 054104.
- [67] H. Werner, P.J. Knowles, G. Knizia, F.R. Manby, M. Schütz, Molpro: a general-purpose quantum chemistry program package, *Wiley Interdiscip. Rev. Comput. Mol. Sci.* 2 (2012) 242–253.
- [68] Y. Zhao, D.G. Truhlar, Exploring the limit of accuracy of the global hybrid meta density functional for main-group thermochemistry, kinetics, and non-covalent interactions, *J. Chem. Theory Comput.* 4 (2008) 1849–1868.
- [69] B.J. Lynch, Y. Zhao, D.G. Truhlar, Effectiveness of diffuse basis functions for calculating relative energies by density functional theory, *J. Phys. Chem. A* 107 (2003) 1384–1388.
- [70] Y. Shao, L.F. Molnar, Y. Jung, J. Kussmann, C. Ochsenfeld, S.T. Brown, et al., Advances in methods and algorithms in a modern quantum chemistry program package, *Phys. Chem. Chem. Phys.* 8 (2006) 3172–3191.
- [71] I.M. Alecu, J. Zheng, Y. Zhao, D.G. Truhlar, Computational thermochemistry: scale factor databases and scale factors for vibrational frequencies obtained from electronic model chemistries, *J. Chem. Theory Comput.* 6 (2010) 2872–2887.
- [72] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, et al., Gaussian 03, Revision B 03, 2004.
- [73] A.D. Boese, J.M.L. Martin, Development of density functionals for thermochemical kinetics, *J. Chem. Phys.* 121 (2004) 3405–3416.
- [74] A.L.L. East, L. Radom, Ab initio statistical thermodynamical models for the computation of third-law entropies, *J. Chem. Phys.* 106 (1997) 6655–6674.
- [75] Cantherm/RMGPpy. <http://greengroup.github.io/RMG-Py> (accessed 18.02.16).
- [76] J.W. Allen, R.W. Ashcraft, G.J. Beran, C.F. Goldsmith, M.R. Harper, A. Jalan, et al., RMG (Reaction Mechanism Generator) version 3.2, 2010. <http://rmg.sourceforge.net/> (accessed 18.02.16).
- [77] C. Eckart, The penetration of a potential barrier by electrons, *Phys. Rev.* 35 (1930) 1303–1309.
- [78] X. Gou, J.A. Miller, W. Sun, Y. Ju, Implementation of PLOG function in Chemkin II and III (2011). <http://engine.princeton.edu/> (accessed 18.02.16).
- [79] A. Chang, J. Bozzelli, A. Dean, Kinetic analysis of complex chemical activation and unimolecular dissociation reactions using QRRK theory and the modified strong collision approximation, *Z Für Phys. Chem. Int. J. Res. Phys. Chem. Chem. Phys.* 214 (2000) 1533.
- [80] J.P. Senosiain, J.A. Miller, The Reaction of n- and i-C4H5 Radicals with Acetylene, *J. Phys. Chem. A* 111 (2007) 3740–3747.
- [81] B.E. Poling, J.M. Prausnitz, J.P. O'Connell, The properties of gases and liquids, McGraw-Hill, New York, U.S., 2001.
- [82] J.R. Welty, C.E. Wicks, G. Rorrer, R.E. Wilson, Fundamentals of momentum, heat, and mass transfer, John Wiley & Sons, 2009.

- [83] M.R. Harper, K.M. Van Geem, S.P. Pyl, G.B. Marin, W.H. Green, Comprehensive reaction mechanism for n-butanol pyrolysis and combustion, *Combust. Flame* 158 (2011) 16–41.
- [84] D.G. Goodwin, H.K. Moffat, R.L. Speth, Cantera: an object-oriented software toolkit for chemical kinetics, thermodynamics, and transport processes, version 2.2.1 (2016) <http://www.cantera.org/> (accessed 18.02.16).
- [85] CHEMKIN-PRO. San Diego: Reaction Design; 2013.
- [86] R. Hanson, D. Davidson, Recent advances in laser absorption and shock tube methods for studies of combustion chemistry, *Prog. Energy. Combust. Sci.* 44 (2014) 103–114.
- [87] R. Hanson, S. Chakraborty, G. Pang, W. Ren, S. Wang, D. Davidson, Constrained reaction volume: a new approach to studying reactive systems in shock tubes, 29th International Symposium on Shock Waves, vol. 1, Springer, 2015, pp. 149–154.
- [88] A.C. Hindmarsh, P.N. Brown, K.E. Grant, S.L. Lee, R. Serban, D.E. Shumaker, C.S. Woodward, SUNDIALS: Suite of nonlinear and differential/algebraic equation solvers, *ACM Trans. Math. Softw.* 31 (2005) 363–396.
- [89] S.C. Chapra, R.P. Canale, Numerical methods for engineers, McGraw-Hill, 2012.
- [90] W.H. Green, Predictive kinetics: a new approach for the 21st Century, *Adv. Chem. Eng.* 32 (2007) 2–50.
- [91] R.G. Susnow, A.M. Dean, W.H. Green, P. Peczak, L.J. Broadbelt, Rate-based construction of kinetic models for complex systems, *J. Phys. Chem. A* 101 (1997) 3731–3740.
- [92] C.W. Gao, J.W. Allen, W.H. Green, R.H. West, Reaction mechanism generator: automatic construction of chemical kinetic mechanisms, *Comput. Phys. Commun.* (2015).
- [93] G. Smith, H. Wang, E.E. Dames, D. Sheen, Foundational Combustion Chemistry, 2013.
- [94] Y. Tao, E.E. Dames, G. Smith, H. Wang, Optimization and uncertainty minimization of the foundational fuel chemistry model, 9th U.S. National Combustion Meeting, Cincinnati, Ohio, 2015.
- [95] D. Healy, H. Curran, S. Dooley, J. Simmie, D. Kalitan, E. Petersen, G. Bourque, Methane/propane mixture oxidation at high pressures and at high, intermediate and low temperatures, *Combust. Flame* 155 (2008) 451–461.
- [96] S. Wang, E.E. Dames, D.F. Davidson, R.K. Hanson, Reaction rate constant of $\text{CH}_2\text{O} + \text{H} = \text{HCO} + \text{H}_2$ revisited: a combined study of direct shock tube measurement and transition state theory calculation, *J. Phys. Chem. A* 118 (2014) 10201–10209.
- [97] V.R. Lambert, R.H. West, Identification, correction, and comparison of detailed kinetic models, 9th U.S. National Combustion Meeting, 2015.
- [98] D.L. Baulch, C.J. Cobos, R.A. Cox, P. Frank, G. Hayman, T. Just, et al., Evaluated kinetic data for combustion modeling. Supplement I, *J. Phys. Chem. Ref. Data* 23 (1994) 847–1033.
- [99] E.E. Dames, D.M. Golden, Master equation modeling of the unimolecular decompositions of hydroxymethyl (CH_2OH) and methoxy (CH_3O) radicals to formaldehyde (CH_2O) + H, *J. Phys. Chem. A* 117 (2013) 7686–7696.
- [100] D.R. Glowacki, C.-H. Liang, C. Morley, M.J. Pilling, S.H. Robertson, MESMER: an open-source master equation solver for multi-energy well reactions, *J. Phys. Chem. A* 116 (2012) 9545–9560.
- [101] S.A. Carr, T.J. Still, M.A. Blitz, A.J. Eskola, M.J. Pilling, P.W. Seakins, R.J. Shannon, B. Wang, S.H. Robertson, Experimental and theoretical study of the kinetics and mechanism of the reaction of OH radicals with dimethyl ether, *J. Phys. Chem. A* 117 (2013) 11142–11154.
- [102] M.E. Jenkin, G.D. Hayman, T.J. Wallington, M.D. Hurley, J.C. Ball, O.J. Nielsen, T. Ellermann, Kinetic and mechanistic study of the self-reaction of methoxymethylperoxy radicals at room temperature, *J. Phys. Chem.* 97 (1993) 11712–11723.
- [103] C.M. Rosado-Reyes, J.S. Francisco, J.J. Szente, M.M. Maricq, L. Frøsig Østergaard, Dimethyl ether oxidation at elevated temperatures (295–600 K), *J. Phys. Chem. A* 109 (2005) 10940–10953.
- [104] P. Dagaut, T.J. Wallington, M.J. Kurylo, Gas phase studies of substituted methylperoxy radicals: the UV absorption spectrum and self-reaction kinetics of $\text{CH}_3\text{OCH}_2\text{O}_2$ —the reaction of CF_2ClO_2 with Cl atoms, *J. Photochem. Photobiol. Chem.* 48 (1989) 187–198.
- [105] X. Song, H. Hou, B. Wang, Mechanistic and kinetic study of the $\text{O} + \text{CHOCH}$ reaction and the unimolecular decomposition of CHOCHO , *Phys. Chem. Chem. Phys.* 7 (2005) 3980–3988.
- [106] S. Dooley, M.P. Burke, M. Chaos, Y. Stein, F.L. Dryer, V.P. Zhukov, O. Finch, Methyl formate oxidation: Speciation data, laminar burning velocities, ignition delay times, and a validated chemical kinetic model, *Int. J. Chem. Kinet.* 42 (2010) 527–549.
- [107] C.K. Westbrook, W.J. Pitz, P.R. Westmoreland, F.L. Dryer, M. Chaos, P. Oßwald, K. Kohse-Höinghaus, T.A. Cool, J. Wang, B. Yang, N. Hansen, T. Kasper, A detailed chemical kinetic reaction mechanism for oxidation of four small alkyl esters in laminar premixed flames, *Proc. Combust. Inst.* 32 (2009) 221–228.
- [108] S. Xu, R. Zhu, M. Lin, Ab initio study of the $\text{OH} + \text{CH}_2\text{O}$ reaction: the effect of the $\text{OH}\cdots\text{OCH}_2$ complex on the H-abstraction kinetics, *Int. J. Chem. Kinet.* 38 (2006) 322–326.
- [109] R. Jensen, S. Korcek, L. Mahoney, M. Zinbo, Liquid-phase autoxidation of organic compounds at elevated temperatures. 1. The stirred flow reactor technique and analysis of primary products from n-hexadecane autoxidation at 120–180 °C, *J. Am. Chem. Soc.* 101 (1979) 7574–7584.
- [110] R. Jensen, S. Korcek, L. Mahoney, M. Zinbo, Liquid-phase autoxidation of organic compounds at elevated temperatures. 2. Kinetics and mechanisms of the formation of cleavage products in n-hexadecane autoxidation, *J. Am. Chem. Soc.* 103 (1981) 1742–1749.
- [111] R. Jensen, M. Zinbo, S. Korcek, HPLC Determination of hydroperoxidic products formed in the autoxidation of n-hexadecane at elevated temperatures, *J. Chromatogr. Sci.* 21 (1983) 394–397.
- [112] R. Jensen, S. Korcek, M. Zinbo, M. Johnson, Initiation in hydrocarbon autoxidation at elevated temperatures, *Int. J. Chem. Kinet.* 22 (1990) 1095–1107.
- [113] K. Moshhammer, A.W. Jasper, D.M. Popolan-Vaida, A. Lucassen, P. Diévert, H. Selim, A.J. Eskola, C.A. Taatjes, S.R. Leone, S.M. Sarathy, Y. Ju, P. Dagaut, K. Kohse-Höinghaus, N. Hansen, Detection and identification of the keto-hydroperoxide ($\text{HOOCH}_2\text{OCHO}$) and other intermediates during low-temperature oxidation of dimethyl ether, *J. Phys. Chem. A* 119 (2015) 7361–7374.
- [114] A. Alan, I.M. Alecu, R. Meana-Pañeda, J. Aguilera-Iparraguirre, K.R. Yang, S.S. Merchant, D.G. Truhlar, W.H. Green, New pathways for formation of acids and carbonyl products in low-temperature oxidation: the Korcek decomposition of γ -ketohydroperoxides, *J. Am. Chem. Soc.* 135 (2013) 11100–11114.
- [115] A.T. Droege, F.P. Tully, Hydrogen-atom abstraction from alkanes by hydroxyl. 3. Propane, *J. Phys. Chem.* 90 (1986) 1949–1954.
- [116] R. Walker, Temperature coefficients for reactions of OH radicals with alkanes between 300 and 1000 K, *Int. J. Chem. Kinet.* 17 (1985) 573–582.
- [117] L.K. Huynh, A. Ratkiewicz, T.N. Truong, Kinetics of the hydrogen abstraction $\text{OH} + \text{Alkane} \rightarrow \text{H}_2\text{O} + \text{Alkyl}$ reaction class: an application of the reaction class transition state theory, *J. Phys. Chem. A* 110 (2006) 473–484.
- [118] R. Sivaramakrishnan, N. Srinivasan, M.-C. Su, J. Michael, High temperature rate constants for OH + alkanes, *Proc. Combust. Inst.* 32 (2009) 107–114.
- [119] J. Badra, E.F. Nasir, A. Farooq, Site-specific rate constant measurements for primary and secondary H- and D-abstraction by OH radicals: propane and n-butane, *J. Phys. Chem. A* 118 (2014) 4652–4660.
- [120] J.D. DeSain, S.J. Klippenstein, J.A. Miller, C.A. Taatjes, Measurements, theory, and modeling of OH formation in ethyl + O and propyl + O reactions, *J. Phys. Chem. A* 108 (2004) 7127–7128.
- [121] J.D. DeSain, S.J. Klippenstein, J.A. Miller, C.A. Taatjes, Measurements, theory, and modeling of OH formation in ethyl + O and propyl + O reactions, *J. Phys. Chem. A* 107 (2003) 4415–4427.
- [122] D.A. Sheen, H. Wang, The method of uncertainty quantification and minimization using polynomial chaos expansions, *Combust. Flame* 158 (2011) 2358–2374.
- [123] F. Pérez, B.E. Granger, IPython: a system for interactive scientific computing, *Comput. Sci. Eng.* 9 (2007) 21–29.